Continuous Random Variable – Introduction

Hello, and welcome to this session on business statistics for entrepreneurs. In the last module, we discussed discrete probability distributions, such as binomial and poisson distributions. In this module, we are going to discuss continuous probability distributions. Our plan will be as follows. First, we are going to look at some fundamentals related to continuous probability distributions. Then, we are going to see examples of continuous probability distributions. Here, we will start with the uniform and the exponential distribution. Then, we will look at the normal distribution, which will be very useful in subsequent modules. Finally, we will look at a few continuous distributions that are related to the normal distribution.

These distributions will be the chi square distribution, the t distribution and the f distribution. Joining me are our bright students, Tejas and Ashwini. As always, they will keep me on track and honest. Tejas and Ashwini, are you ready to start? Yes, I am. Yes, is this going to be similar to what we did for discrete distributions? By and large, with some important differences. So let's get started. First, let's refresh our memory about some examples of discrete random variables. We know discrete random variables can take specific integer values. Let's say you were making sales calls to five potential customers. The number of potential customers that then end up placing an order as a result of the phone call could be a random variable that takes values from zero to five.

As another example, Tejas, I think you mentioned this last time. If we were to inspect a batch of 50 radios received from a supplier, then the number of defective radios in this batch would be a discrete random variable, taking values between 0 and 50. Yes, I remember. And Ashwini, I think you had one too, related to the number of customers in a restaurant, right?

Yes. Yes, the number of customers that arrive in a day could be anywhere from zero to a very large number. Yes, and all of these are discrete random variables. Continuous random variables take fractional values. For example, the time between two consecutive arrivals of customers to a bank. This can take any positive value, not just integer values.

Also, if you looked at the amount of fluid in a soft drink bottle, suppose the bottle is labeled 750 ml. The bottle in reality might have a quantity somewhere between, let's say, 740 ml and

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760 ml. Third, as an example, consider the case where the ideal baking temperature of pizza, the actual oven temperature, may have a value between, let's say, 275 degrees Celsius and 320 degrees Celsius.

All of these are examples of continuous random variables. They take values in a range sometimes a limited range, sometimes an infinite range. Like with discrete random variables, for continuous random variables too, we are interested in some very specific questions. But we always denote the random variable itself by a capital letter, usually capital X, or capital Y, or capital Z.

And then we denote the particular value that a random variable can take, by the corresponding small letter, small x, small y, or small z. Then, we have the probability density function, small f of x, which is defined for the values that the random variable can take. However, for continuous random variables, It has a different interpretation than what we had for discrete random variables.

We'll see that shortly. Like we had for discrete random variables, we have the cumulative distribution function, capital F of x. We also have the expected value, E of x, and the variance of x, and so on. Let's discuss each of these now in more detail. To compare discrete and continuous random variables, let's look at two examples.

The first example is the one we saw earlier. Tejas, if you recall, we had looked at a problem that Kedar Apte, the quality manager at Sangli, had. He had collected data related to the number of defective pumps in a production per day. We had a discrete probability distribution. Do you remember that? Yes, I remember.

The number of defects were ranging from zero to four with specific probabilities for each. Yes, exactly, Tejas. And you see the distribution in the picture here. The probabilities corresponding to 0, 1, 2, 3, and 4 defects are 0.4, 0.25, 0.2, 0.05, and 0.1 respectively. The second example we are going to see is related to a continuous random variable.

Megajane, located in Surat, has collected data about waiting times that she experiences when she calls a customer service desk. These waiting times, of course, can take any positive value.

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You can see the function small f of x plotted here. Notice that since waiting times can take fractional values, Small f of x is defined for all positive values of x, that is x greater or equal to zero.

A fundamental difference that separates the discrete and continuous random variables is in terms of how probabilities are computed. For discrete random variables, we talked about the probability mass function small f of x. This is a function that assigns probabilities to the individual values that a random variable can take.

So, in Kedar Apte's case, The small f of x gave us the probabilities of having a zero defect, one defect, and so on. With a continuous random variable, the counterpart of the probability function is the probability density function. It is also denoted by small f of x. However, the difference is that the probability density function does not directly provide the probabilities.

Instead, the area under the graph of small f of x, corresponding to a given interval is what provides the probability that the continuous random variable capital X assumes a value in that interval. So, when we compute the probabilities for a continuous random variable, we are always computing the probability that the random variable assumes a value in a given interval.

So, probability is no longer defined as the value of f of x for a given x. But the area under small f of x for an interval of values that x can take. Yes, that's right, Ashwini. Now, Tejas, what would be the area under the curve for a particular point? Area under a curve for a particular point would be zero. Is that right? Yes, absolutely. So, because the area under the curve of small f of x at any particular point would be equal to zero. One of the implications of this definition of probability for a continuous random variable is that the probability of any particular value of the random variable is equal to zero.

Now, despite these differences, there are a lot of similarities between the small f of x for discrete random variables. And the small f of x for continuous random variables. The probability mass function small f of x for a discrete random variable can only take positive values or 0, since they correspond to probabilities directly. And the sum of small f of x, the probability mass function over all values that the random variable can take comes out to be equal to 1. For continuous random variables, the probability density function small f of x can

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also take only positive values or zero. And if you integrate the function small f of x over all values that the random variable capital X can take, we would get this integral to be equal to one.

In other words, for a continuous random variable, the overall area under the curve corresponding to small f of x is always equal to 1. Further, for a discrete random variable, if we wanted the probability that the random variable took values between sum a and b, In that case, we just added up the probabilities given by small f of x for all values that the random variable could take between a and b inclusive.

Now, if capital X is a continuous random variable and we want the probability that it takes values between two numbers a and b, then instead of adding, we would integrate the function small f of x dx over all values of x ranging from a to b. The value of this integral would correspond to the required probability.

So, in Kedar Apte's case, the probability of number of defect being between 1 and 3, both inclusive, was 0. 25 plus 0. 2 plus 0. 05. which was equal to 0. 50. Yes, that's right. But I hate integration. Why? I like it. It is fun. Well, whether you like or whether you hate integration, our focus will be more on the end result of integration rather than the process of integration. So you almost will never have to do integration from first principles. We will either use a software like Excel or use tables to derive the end result. We just need to understand that the role the integration plays in continuous random variables is similar to the role the summation plays in discrete random variables.

Next, let's talk about the cumulative distribution function capital F of x. For a discrete random variable, the capital F of x provides the probability that the random variable capital X takes values equal to small x or lower. So in Kedar Apte's example, if you wanted the probability of having three or fewer defective pumps in a day, we would add the probability of having three, two, one, or zero defects in a given day.

In general, if you wanted the probability of capital X, taking values less than or equal to some value small a, we would add the values of small f of x from the minimum value all the way up

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to and including small a. For a continuous random variable, we also define a cumulative distribution function. We denote it by the same letter capital F of x.

In this case too, it provides a probability that the continuous random variable capital X takes values of small x or lower. However, the way in which we compute this probability is not by summation. Instead, we integrate small f of x dx over all values that the random variable can take from its minimum all the way and up to including small x.

Therefore, if you wanted to calculate the probability of a random variable capital X taking values less than or equal to a, we would denote that for the continuous random variable by capital F of a, and we would compute it by integrating small f of x from the minimum value it can take all the way up to a.

Now, let's talk about the expectation and the variance of the random variable capital X. If it was a discrete random variable, then to calculate its expectation E of X, we would have done the following. For each value of the random variable small x, we would have multiplied small x by the probability mass function small f of x.

And then added this product over all values that the random variable capital X can take. To compute the variance of the random variable, var of x, we would have done the following. For each value of the random variable, we would have computed x minus E of x squared. The squared deviation from the mean or the expectation.

Then we would multiply the square deviation by the probability mass function small f of x. And finally, we would have computed the sum of these products for every value that the random variable can take. And that would compute the var of x or the variance of x. How does this change if x was a continuous random variable?

If x is a continuous random variable, then to compute the expectation of x, we would do the following. For each value of the random variable, small x, we would have multiplied small x by its probability density function, small f of x, and then integrate this product over all values that the random variable capital X can take to compute the variance of the random variable, var of x, like in the discrete case, we would do the following.

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For each value of the random variable, we would have computed x minus E of x square, the square deviation from the mean or the expectation. Then we would multiply the square deviation by the probability density function, small f of x. And finally, we will compute not the sum of these products, but the integral of the product over all values that the random variable can take.

And we would compute the var of x to be equal to the value of this integral. This is similar to the discrete case, except for the integrals part. True. So to summarize, the definitions and properties of continuous random variables For a continuous random variable capital X, we have the probability density function or pdf of capital X denoted by small f of x.

This function, small f of x, satisfies the condition that small f of x is always 0 or greater. And if you integrate the value of small f of x times dx over all values that the random variable can take, this integral comes out to be 1. The cumulative distribution function CDF of the random variable, capital X, is denoted by capital F of x.

And this provides us the probability that the random variable capital X takes a value of small x or lower. The tail probability denoted by f bar of x provides us the probability that the random variable capital X takes values greater than small x. And that is equals to 1 minus capital F of x. Yes, that is correct.

And the expectation of the random variable E of x is given by the integral over all values that the random variable can take of the product of x times f of x dx. The variance of the random variable var of x is defined as the integral taken over all values that the random variable can take of the product of the square deviations x minus E of x square and the small f of x, the probability density function of x. Now that we have seen the fundamentals, let's see them in the action in the context of some commonly used continuous random variables. We will start by looking at the uniform random variable and the exponential random variable.

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Uniform Distribution

So, what is the uniform random variable and what is its distribution? If a random variable can take any value between two limits, a lower limit capital L and an upper limit capital U, and each value is equally likely to happen, then this kind of random variable is a uniform random variable. The probability density function, small f of x of this random variable is known as the uniform distribution.

In other words, whenever the probability is proportional to the length of the interval, the random variable is uniformly distributed. Now, for a uniform random variable, the probability density function, small f of x is defined as 1 divided by the upper limit u minus the lower limit L. Note that small f of x is defined for all values between the lower limit capital L and the upper limit capital U.

But small f of x is equal to 0 elsewhere. The cumulative distribution function capital F of x, which is obtained by the integration as we just discussed, would come out to be x minus L divided by U minus L. The tail probability F bar of x would then be 1 minus the capital F of x, and that would come out to be U minus x divided by U minus L.

Then we can apply the definitions we saw a little while ago to compute the probability that a uniform random variable capital X takes values between a and b. And that would be given by b minus a divided by u minus L. And if we apply the definitions of expectation and variance on this probability density function, we would get the expectation of x to be equal to L plus U divided by 2. That is the average of the lower and upper limits. Yes, that's right. And the variance of x obtained by computing the integral of the square deviations multiplied by the probabilities would come out to be equal to U minus L squared divided by 12.

Let's look at an example now. Nilesh Shah is a regular business traveler who travels from Vadodara to Ahmedabad for work. According to him, the average driving time is around 120 minutes. But usually, when it takes a ride, it takes anywhere between 100 minutes and 140 minutes. We could then model this driving time as a uniformly distributed random variable that takes values between the lower limit 100 minutes and the upper limit 140 minutes.

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Using this information, we can define the probability density function. small f of x and use that to compute several useful probabilities. For example, what would be the probability that the drive takes 130 minutes or less? Second, what would be the probability that the drive takes more than 105 minutes? And third, what is the probability that the drive takes exactly 120 minutes?

If x denotes the driving time and it's uniformly distributed between the lower limit of 100 minutes and upper limit of 140 minutes, what would be the expectation, Tejas? We know that the expectation is equal to the average of the lower and upper limits, which would be 100 plus 140 divided by 2, which would be 120 minutes.

Yes, and Nilesh sir wants to know the probability that the drive would take 130 minutes or less. So, Ashwini, how would we find that? We have to compute the CDF for 130 or capital F of 130. Okay. Yes, and that would be given by 130 minus 100 divided by 140 minus 100 which would come out to be 0. 75.

Next, Nilesh Shah wants to know the probability that the drive would take more than 105 minutes. Tejas, how would we tackle this one? In this case, we want the tail probabilities F bar for 105. Yes, and that would be given by 140 minus 105 divided by 140 minus 100, which would come out to be 0. 875. So therefore, there would be an 87. 5 percent chance that the drive will take more than 105 minutes.

Third, Nilesh wants to know what is the probability that the drive takes exactly 120 minutes. Tejas, how would you do this? We need to compute the probability of exactly 120 minutes and that would be capital F of 120. That would be 1 divided by the upper limit minus lower limit, which should be 1 divided by 140 minus 100, which would be 0.025. No, no, no, no, no, no. That's not true. If you recall, I said that for a continuous random variable, we define probability for values that a random variable can take in an interval. And because it is defined as the area under the PDF for the corresponding interval, an area under the curve at a point is going to be equal to 0.

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So the probability of a continuous random variable taking a particular value would always be equal to 0. So in this case, the probability of the drive taking a value exactly equal to 120 minutes is going to be equal to zero. Does that mean the probability that the drive takes 110 minutes would be equal to zero?

130 minutes is equal to zero. Is this true for any value and not just 120? Yes, that's right. The probability that a random variable will take any single value would be equal to zero if it is a continuous random variable. But we can always determine the probability of a continuous random variable taking values in an interval, but not at a point.

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Examples of Uniform Distribution

Okay, let's look at another example. Meena Desai, living in Gandhinagar, has just purchased a new Samsung Galaxy phone. She was told that the battery will last for an average of 10 hours and she will need charging only after that. As she is using the phone, she notices that the battery requires charging, usually sometime between 8 to 12 hours of usage.

She has some relevant questions for us. First, She wants to know what is the probability that a recharge will be needed after at most nine hours of usage. Second, she wants to know what is the probability that the battery will last for at least 11 hours before a recharge is needed. Third, she wants to know the 80th percentile of the time to recharge her new phone? Let's see if we can answer these questions.

From her description, it is clear that the time before the phone needs a recharge is a random variable. Let us denote it by capital X. We know that capital X varies between 8 hours and 12 hours. Therefore, we can model the time of usage until the phone needs charging as a uniform random variable that has a lower limit of 8 hours and an upper limit of 12 hours. First, she wants to know the probability that the recharge will be needed after at most 9 hours. Ashwini, how will you compute this? We have the lower limit to be equals to L equals to 8R upper limit U equals to 12 hours. We want the probability that the random variable X takes a value less than or equals to 9. So we want capital F of 9. That's right. And that would be equal to 9 minus 8 divided by 12 minus 8, which would be 0. 25. That is, there is a 25 percent chance that the phone will require recharge within 9 hours. Next, she wants to know what is the probability that the battery will last for at least 11 hours before a recharge is needed.

Tejas, what should we calculate here? In this case, we want the probability of capital X is greater or equal to 11. So we need the tail probability capital F bar of 11. That's right. And that would be given by 12 minus the lower limit 11 divided by 12 minus 8 which is L minus U. This also comes out to be 0.25. So there is a 25 percent chance that the battery will last for at least 11 hours before a recharge is needed.

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Third, she wants to know what is the 80th percentile of the time to recharge the phone. What is 80th percentile of random variable X? We only saw percentiles with respect to data. How do you compute percentiles with respect to random variables?

The concept of percentile can be easily applied to random variables. Here, the 80th percentile refers to a probability of 0. 8. What Meena wants to know is the value of small x such that the probability of capital X is less than or equal to small x is equal to 0. 8. Oh, just the inverse calculation involving cumulative distributions?

Yes, that's right. So we want the value of small x such that small x minus 8 divided by 12 minus 8 is equal to 0. 8. Solving for x, we get small x equal to 11. 2 hours. This means that there is an 80 percent chance that her phone will require charging within 11. 2 hours. Uniform distribution is easy.

Yes, it is both conceptually and in terms of calculation an easy one. Do you know that there is a discrete version of the uniform distribution? Is there? But we only did the Bernoulli, binomial and Poisson. No, we did the discrete version of uniform too. We did. I don't remember. I'll give you a hint. When we did it, you mentioned that you did not like it.

Don't remember? It was the example of the unbiased die, the probability of it taking discrete values 1, 2, 3, all the way up to 6 with uniform probability of 1 over 6. Remember? You said then, why do we always talk about coins and dice when we talk about probability? Yes, I remember. Discrete version of uniform is actually easy too.

Yes, they are all hard until you understand them. But once you understand a concept, it's always easy.

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Exponential Distribution

Let us look at another continuous random variable known as the exponential random variable. We will see its probability density function, small f of x. Discuss some of its properties, and then see some examples. Suppose an event of interest occurs in a random manner, but at an average rate or frequency of lambda per hour, that is constant.

Then, the time between two such random events of interest would be a random variable. Let us denote this random variable by capital X. The capital X would have an average of 1 over lambda hours, and in many cases, this random time between two events can be modeled as an exponential random variable. The probability density function, small f of x, of this random variable is called as the exponential distribution.

For an exponential distribution, the probability density function, small f of x, is given by lambda times e raised to minus lambda times x. Notice that the exponential random variable corresponds to the time between the occurrences of random events. It is not counting events, but it is measuring the time between events.

Therefore, this time, the random variable, can only take positive values. But it can take large positive values. So, it is defined for all x greater or equal to zero. Now, using the probability density function small f of x, we can compute the cumulative distribution function capital F of x and the tail probability f bar of x.

The cumulative distribution function capital F of x for any value of small x comes out to be equal to 1 minus e raised to minus lambda times small x. How do we get this cumulative distribution function? If you recall, I had said that in order to get the cumulative distribution function for continuous random variables, we would need to integrate over an interval.

Yes, I remember. Yes, so now if you take the probability density function small f of x, and if you integrate it over all values starting from the least value of x, which is equal to 0, and all the way up to small x that we are interested in, then the result of that integration would be 1 minus e raised to minus lambda times x.

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Now I remember, you had mentioned about integration. Yes, and I had also said that we will not focus on the mechanics of integration, but more on the end result obtained through the integration. Yes, I remember. Therefore, all we are concerned is that the cumulative distribution function capital F of x is going to be equal to 1 minus e raised to minus lambda times small x.

Next, the tail probability F bar of x is the probability that the random variable capital X takes a value greater than small x. Again, if we integrate the probability density function, from the value of small x all the way up to infinity, we will get e raised to minus lambda times small x. You can also see that f bar of x is simply obtained by taking 1 minus capital F of x.

Yes, we see that. Next, suppose you want the probability that the random variable take capital X takes a value between a and b. Then we can use the cumulative distribution function and tail probabilities to compute this. We can write the probability that the random variable lies between a and b to be equal to the probability that the random variable capital X takes a value greater than small a, minus the probability that the random variable capital X takes a value greater than b.

In all of those cases, the random variable would lie between a and b. And mathematically, that would come out to be, e raised to minus lambda times a minus e raised to minus lambda times b. Finally, we can compute the expectation of the exponential random variable and its variance. To compute the expectation of the exponential random variable, we can integrate x times f of x over all values that x can take, which in this case would be zero to infinity.

And if we did that, the outcome is pretty simple. It comes out to be one over lambda. The variance of the exponential random variable can be obtained by integrating the square deviations from the mean that we just calculated multiplied by the PDF. Notice that the range for this integration would also be over all values of x, which is from 0 to infinity. And if you did that, the outcome would come to be 1 over lambda square.

Now, Tejas, you said you enjoyed integration. So, I suggest you can try verifying both expressions for the mean and the variance of the exponential random variable and the as an exercise in calculus for your own benefit. Okay, I will do it. It does not seem hard. I don't want to try integration if I can avoid it. Well, then Ashwini, I have another task for you. Compare

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the expectation and the standard deviations of x and tell me if you notice anything peculiar. The expectation of X is 1 over lambda, and the variance is 1 over lambda square.

That's right. So what would be the standard deviation of the exponential random variable? It's the square root of the variance, which also comes to be 1 by lambda, the mean. Interesting. Yes, you will notice. that for the exponential distribution, it has a unique property where the mean and the standard deviation are the same.Wasn't the mean and standard deviation the same for the positive distribution? Was it? Let me check. No mean and variance were the same. Equal to capital lambda. Yes, for the Poisson, the mean and the variance were both equal and were denoted by capital lambda. Are the small lambda of exponential and capital lambda of Poisson connected?

Well, you both will make very good detectives. Yes, they are connected and we will see that connection in a bit. Let's look at some properties of the exponential random variable and then some applications before we look at some examples. We saw that the exponential random variable can only take positive values.

since it corresponds to the time between some random events. We also saw that its mean and standard deviations are equal. If you look at the plot of the probability density function or PDF, you will notice that the exponential distribution is skewed to the right. It is not symmetric about the average. Also, the exponential distribution has a special property known as the memoryless property. We will see that in more detail a little bit later. But now, what are some applications of the exponential random variable? We know that it models the time between random occurrences of events. One of the common applications is to use it to model the time taken to complete manual tasks.

For example, the time to manually assemble a product, or the time duration of a phone call answered by an agent at a call center. The exponential distribution is also used to model the lifetime of equipment, such as a bulb or any other electronic equipment. The exponential random variable is also used to model the time it takes before a component, let's say in an automobile or an aerospace engine, requires repair or maintenance. The exponential random variable is also used to model the waiting time of a customer standing in queue for service.

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It seems it has a lot of applications. Yes. So now let's look at a few examples. Megha Jain in Surat manages travel bookings for her company. A frustrating part of her job is the waiting time that she experiences when she calls the service desk for travel bookings.

You mean the time we are made to listen to music or promotions while waiting for an agent to pick up the phone? Yes, the time when they say, your call is important to us, but then they make you wait for a long time. Yes, that's the waiting time we are modeling as an exponential random variable. She has collected data on these waiting times, and they have an average of two minutes.

She would like answers to several probability related questions. First, she wants to know what is the probability that she waits for less than one minute for the service desk to respond to her call. Second, she wants to know what is the probability that she waits for more than three minutes for the service desk to respond to her call. Third, she wants to know what is the probability that she waits between one and three minutes for the service desk to respond to her account. And fourth, she wants to know what is the 90th percentile of her waiting time. Shall we try and answer this? Before we do that, I would like to talk a little bit about how we can compute these probabilities using Excel.

Similar to what we did for the binomial distribution, we can use some of the inbuilt functions in Excel to do similar calculations for the exponential distribution. First, we open a new spreadsheet in Microsoft Excel, and then we use the inbuilt command for the exponential distribution called =expand. dist( ) to create a table that allows us to compute various probabilities. In Excel, the function =expand. dist( ) has three arguments. The first argument is small x, the value for which we are computing the probabilities. The second argument is lambda, the rate of events. And the third argument is a toggle, false or true, where selecting false gives us the probability density function small f of x corresponding to the other parameters and true gives us the cumulative distribution function capital F of x corresponding to the other parameters selected. Let's now see this in a template. Here in cell B3, I have entered the value of small lambda. In this case, it is equal to 0. 2.

Then in cell 3, I have calculated the mean of the exponential as 1 over lambda equal to 5 in this case. In cell D3, I have computed the standard deviation, which is equal to the mean for the

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exponential. Next, there are two sections below these inputs. The first section computes the probability distributions.

The other section can be used to do the inverse calculations. In both sections, you can enter values in the green cells and see the outputs in the white cells. In the first section, by entering the value of small x, you can compute the cumulative distribution, probability of capital X less than or equal to some small x, or the tail probabilities, F bar of small x, which is the probability that capital X is greater than small x, and also the probability that small x takes values between sum a and sum b.

In the section below used for inverse calculations, you can find the value of small x if you are given the value of either the cumulative probability, the probability of capital X being less than or equal to small x, or the tail probabilities f bar of small x, which is the probability of capital X being greater than or equal to small x.

This is nice. We will use the inverse section for the question related to percentiles, right? Yes, but wait, there is more. On the right hand side, we have a table in columns P, Q and R that computes the PDF, the CDF for various values of small x and for the value of lambda specified in cell B3. You can see the related plot of the PDF as well.

Now, let's answer Megha Jain's questions. We know she does not like waiting. The first question she wants answered is what is the probability that she waits for less than one minute for the service desk to respond to her call? We modeled her waiting time for the service desk to respond as an exponential random variable.We know that it has a mean of 2. So Ashwini What would be the parameter lambda in this case? The mean of the exponential is 1 over lambda, when lambda is the parameter. Yes, and since the mean is 2, the parameter lambda would be 0. 5. Now, we want the probability that the random variable capital X takes the value of 1 minute or less.

That is, we want capital F of x which would be given by 1 minus e raised to minus lambda times x, where lambda is 0.5 and small x is equal to one. So that would be one minus erase to the power minus half, and that comes out to be 0.393. We could also use the template to compute this. We enter the value of lambda in cell B3. We notice that the mean is two minutes

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in cell C3, so we know that we have done that correctly. And then in the cumulative probability section, we enter the value of small x equal to 1, and we read out the probability to be equal to 0. 393.

Next, Megha wants to know, what is the probability that she waits for more than three minutes for the service desk to respond to her call. Tejas, can you help with this one? We have lambda which is equal to 0. 5, as Ashwini just computed. In this case, we want the tail probability, f bar of 3. And we want the probability that x is greater than 3. That would be e raised to minus lambda times 3. Which is e raised to minus 1. 5, which would come out to be 0. 223. We could also use the template to calculate this.

We go to the tail probability section in the top half, and we enter 3, and we read off the probability to be equal to 0.223. Next, Megha wants to know what is the probability that she waits between 1 and 3 minutes to before the service desk responds to her call. Ashwini, help us out with this one. Lambda is 0. 5 and we want the probability of the random variable x takes values between 1 and 3. Yes, and that would be e raised to minus lambda times a minus e raised to minus lambda times b, where a is equal to 1 and b is equal to 3. And that would come out to be 0. 383. We can also go to the template and compute this probability. We use the top half again, and put 1 and 3 as the values of a and b, and then we can get the probability to be equal to 0. 383 Next mega wants to know the 90th percentile of the waiting time. In this case, we want to find that value of small x such that the probability of the random variable capital X is less than or equal to small x is equal to 90 percent. We use the cumulative distribution function now.

We want e raised to minus lambda times x to be equal to 0. 9. Ashwini has told us that lambda is equal to 0. 5. Then, solving for x, We get small x to be equal to 4. 605. In other words, there is a 90 percent chance that Mega's waiting time would be less than or equal to 4. 605 minutes. We can also compute this using the template.To do this now, we have an inverse calculation. So we use the bottom half of the template. We input the probability given to be 0. 9. And we find out that the value of x that gives us the cumulative probability to be equal to 0. 9 is equal to 4. 60. That wasn't hard, was it? No, definitely not. The templates definitely make it easier. And the equations themselves are not complicated.

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Memoryless Property of the Exponential Distribution

Now that you have got a feel for how the exponential random variables work, we will see an important property of the exponential random variable known as the memoryless property. Let us look at the memoryless property of the exponential random variable. Assume that the time to some event is random and let us denote it by capital X, and let it have an exponential distribution with mean equal to 1 over lambda. Then, conditional on the event that capital X, the time to an event, is already greater than some value at 0, we define another random variable capital Y, corresponding to the remaining time for the event to occur.

That is, the random variable Y is equal to capital X minus at 0. Now, the memoryless property states that this remaining time, capital Y, also has the same exponential distribution with mean equal to 1 over lambda. In other words, If the event hasn't occurred by time at 0, the additional time for the event to occur beyond at 0, which is y, is equal to x minus at 0 as the same exponential distribution as x.

In simple terms, it appears as if the process just seems to forget, that a time at 0 has already elapsed and it acts as though it is just starting from fresh at time 0. The fact that there is already some time at 0 has elapsed does not change the distribution of the additional time that has to elapse before an event occurs.

Mathematically, this translates to the following. Since x has an exponential distribution with parameter lambda and mean equal to 1 over lambda, the probability that capital X is more than some value small x is given by the tail probability f bar of x equal to e raised to minus lambda times x. Now, if capital Y is a random variable that represents the additional time beyond some at 0 that has elapsed before the event occurs, then the probability that capital Y is greater than some value small y is going to be equal to the probability that capital X is greater than at 0 plus y conditional on the fact that capital X is already greater than at 0.

Now, we can write out this conditional probability, and we will get the probability that capital Y is greater than small y to be equal to e raised to minus lambda times x zero plus small y divided by e raised to minus lambda times at zero, which would simplify to e raised to minus lambda times y.

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Therefore, we get the probability that the additional time beyond at zero for the event to occur, denoted by the random variable capital Y would take a value greater than some value small y is equal to e raised to minus lambda times y. Notice that this is the same as the tail probability distribution for our original random variable capital X.

This implies that the random variable corresponding to the additional time capital Y and our original random variable capital X have the same exponential distribution, thus illustrating the memoryless property. Shall we see an example? On the first of every month, Shanti Banshaah catches a bus to go to the main post office in Anand to withdraw her pension credited to her account.

She lives on a bus route where the arrivals of bus is infrequent and therefore she has noticed that the time taken for her to wait for a specific bus that is needed at her bus stop is random and has an average of 20 minutes. She would like some help from us to manage her waiting times better.

She wants us to answer several questions. First, she would like to know what is the probability that she spends 15 minutes or less waiting for the bus. Second, she would like to know what is the probability that she spends more than 30 minutes waiting for the bus. Third, she would like to know what is the probability that she spends between 15 and 30 minutes waiting for the bus.

Fourth, on a day where she has already waited for 10 minutes, what is the probability that she must wait for another 10 minutes or more for the bus? And finally, on a day where she has already waited 20 minutes for the bus, what is the probability that she has to wait another 10 minutes or more for the bus?

Shall we answer these questions? Yes, let's help Shanti bahan. I feel sorry for her already. We know that Shanti bahan waits on an average 20 minutes for the bus. And since we don't have any additional information about this waiting time, we can model the random variable corresponding to this waiting time to be equal to an exponential random variable with mean of 20 minutes.

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So Ashwini, what would be the lambda in this case? In this case, lambda would be 1 over 20, which would be 0.05. Yes, that is correct. Now, first Shanti bahan wants to know what is the probability that she spends 15 minutes or less waiting for the bus. Tejas, what would that be? We would need to calculate the cumulative probability f of 15, right?

Yes, that's correct. And we would want to calculate the probability that the random variable capital X takes a value of 15 minutes or less, which would be equal to 1 minus e raised to minus 0. 05 times 15, and that comes out to 0. 528. We can also use the template to compute this. We put in the value of lambda to be equal to 0. 05 in cell B3. Then the average gets updated to 20 minutes. Then we go to the probability section in the top, and in the cumulative probability part, we put in the value of small x to be equal to 15. And we read out the probability to be equal to 0. 528. So there is a 52 percent chance that Shanti bahan will wait for 15 minutes or less.

That makes me feel better. What's the next question? Next, Shanti Ben wants to know the probability that she spends more than 30 minutes waiting for the bus. Ashwini, how do we go about this one? We need the tail probability, F bar, corresponding to 30. Yes, that is right. And we compute it as the probability of X being greater or equal to 30 to be equal to e raised to minus lambda times 30, where lambda is equal to 0. 05. Computing that gives us the probability to be equal to 0. 223. We can also do this on the template. We already have put in the value of lambda to be equal to 0.05. We go to the upper part and look at the tail probability section and enter the value of 30 to get the tail probability to be equal to 0.223.

Next, she wants to know the probability that she spends between 15 and 30 minutes waiting for the bus. Ashwini, help us out. In this case, we want the probability that capital X lies between values a and b, where a is 15 and b is 30. That's right. And so can you compute that Tejas? The probability of capital X taking values between 15 and 30 would be equal to e raised to minus lambda times 15 minus e raised to minus lambda times 30. Yes, and that should come out to be equal to 0. 249. Now, if I put that in the template, I should get a probability of 0. 249, and that's roughly a 25 percent chance that she waits between 15 to 30 minutes for her bus.

Next, she wants to know that if she has already waited for 10 minutes for the bus, What is the probability that she will need to wait another 10 minutes or more for the bus to arrive? So in

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this case, we know that the waiting time denoted by capital X is already greater than 10 minutes. Conditional on this, we want the probability that the waiting time will be at least 10 more minutes. In other words, we want the probability that X is more than 20 minutes, given that X is greater than 10 minutes.

This probability would be equal to e raised to minus lambda times 20 divided by e raised to minus lambda times 10, which would come out to be e raised to minus lambda times 10, which is 0. 607, a roughly 60 percent chance that she would have to wait another 10 minutes, given that she has already waited for 10 minutes. The probability of waiting for more than 10 minutes did not change based on the fact that she already waited for 10 minutes. The exponential random variable conveniently forgot how long Shanti bahan has waited. Exactly.

Now, let's see a small variation of this. Now, let's say that she has already waited for 20 minutes for the bus. Now, what is the probability that she will need to wait another 10 minutes or more for the bus to arrive? Unlike in the case that we just saw, in this case, we know that the waiting time denoted by capital X is already greater than 20 minutes. Conditional on this, we want the probability that the waiting time will be at least 10 more minutes.

In other words, we want the probability that capital X is more than 30 minutes, given that capital X is greater than 20 minutes. This probability would be equal to e raised to minus lambda times 30 divided by e raised to minus lambda times 20. Which would come out to be e raised to minus lambda times 10 again, which is 0.607 again. Indicating that roughly a 60 percent chance that Shanti bahan would have to wait for another 10 minutes, given that she has already waited for 20 minutes in this case. The probability of waiting for more than 10 minutes did not change based on the fact that she already waited for 20 minutes.

What a strange property! The exponential random variable conveniently forgot how long Shanti bahan has waited. Poor Shantibain! Let's call a taxi for her.

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Relation Between Exponential & Poisson Distribution

Ashwini, if you recall, when we did the Poisson random variable and its distribution, we discussed the Poisson process. Yes, it was counting random events occurring over time. Yes, the Poisson process is a particular example of what is known as a stochastic process. A Poisson process can be thought of as a series of random events occurring over time, where the time intervals between these events are independent random variables, each having the same distribution known as the exponential distribution.

So the Poisson process and exponential distribution are related. If the time between random events is exponentially distributed, the counting process is Poisson. Yes, you both latched on to a very interesting connection. In fact, there is always a connection between the time between arrivals and the counting process.

Yes, you showed us a picture with dots and gaps between dots earlier. The number of dots occurring up to time t corresponds to the counting process. And the gap between the dots corresponds to the continuous random variable. Exactly. You remember the figure. I am impressed. We pay close attention to what you say.

Good for you. Now, if the gap between the dots is deterministic, no randomness, Say an event occurs every 10 minutes, precisely, no randomness, then if I asked you, how many events will occur in 60 minutes? It would be 60 divided by 10, equal to 6. Yes, equal to 6. Exactly. Not more, not less. Now, if the event occurs on an average every 10 minutes, and the time between the events is exponential, right?

I'm just adding some randomness. But keeping the average time between the events to be the same, 10 minutes. Then, on an average, how many events would occur in 60 minutes? Sometime it will be more than 10. Sometime less than 10 minutes between events on an average, it is 10 minutes between events. So I would see 6 events on an average.

True. But are you sure you will always see exactly 6 events in 60 minutes in this case? Definitely not. Now the numbers of events in 60 minutes is random. Precisely, and the number of events in 60 minutes would be a Poisson process with an average equal to capital lambda

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equal to 6 events. If the gap between events is deterministic, the counting process is deterministic. If the gap between events is exponential, the counting process is a Poisson process. You summarized that very nicely, Tejas. I couldn't have said it better.

Can we see an example related to this connection? Yes, let's revisit an old example. Remember Shivali Patil in Solapur, the chief officer at the PWD? Overseeing the Maharashtra State Highway 3 between Solapur and Nanded? Yes, she was overseeing repair work. Yes, and the number of defects in a one kilometer stretch was a random variable. We assume the occurrence of defects to be a stochastic process, to be a Poisson process. Yes, exactly. With an average of three defects in one kilometer. That is, capital lambda is equal to three. Now, what is the distribution of the distance between two defects? If the occurrence is a Poisson process, the distance between two defects must be an exponential distribution. Correct. And what is the average distance between two defects on the highway? I am only asking the average, because we know defects occur randomly.

On an average, three defects in one kilometer. So, on an average, one by three kilometer between defects. Yes, averages are easy to figure out. Now, if the distance between two defects on the highway has an exponential distribution with an average of 0. 33 kilometers, What is the probability that you see at least half a kilometer stretch of the highway with no defects?

The distance between or the exponential random variable must take a value more than 0. 5 when the average is 0. 333. Yes Tejas and what does the template tell us? We need the tail probability corresponding to 0.5 when small lambda for the exponential is 3. The template says that it would be 0. 223. Yes, that is correct.

And what is the probability that we go at least a 2 kilometer stretch without any defects? It should be smaller than 0. 223. And what is it? Tail probability corresponding to 2, which is 0.002. Expect a bumpy ride. Shivali is not going to be happy. Yes, but you see how we are able to switch between the Poisson and the exponential.

Yes, it was there in front of us all the time. We were solving problems about poison. We just did not realize it. Yes, shall we do one more example? Yes, let us do one more example.

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Examples

Rohit Dave runs a garage shop in Bhavnagar. To boost sales on weekends, Rohit announces that his staff can guarantee completing the oil change on a car in 20 minutes. If it takes more than 20 minutes for the oil change, his garage will provide a 50 percent discount. How risky is this promise? Seems like a bold offer to me. We will find out. We can use our knowledge of the exponential distribution to answer some questions for him. First, he wants to know what is the probability that Rohit's team will take less than 20 minutes to do an oil change. Second, what is the probability that Rohit's team will take between 15 and 30 minutes to do an oil change?

Third, what is the probability that Rohit's team does more than 15 oil changes in a four hour morning shift? And fourth, what is the probability that Rohit's team does more than 30 oil changes during an eight hour day? Let's try and answer these questions.

We know that the time to do an oil change is a random variable. Since we have only information about the average, We can assume that the time taken to do an oil change is exponentially distributed with the average of 20 minutes. This implies that the lambda for the exponential distribution is equal to 1 over 20, which is 0. 05. Then Ashwini, how do we calculate the probability that Rohit's team will take less than 20 minutes to do an oil change?

We just found a cumulative probability, capital F of 20, right? Yes, that is correct. So we update the value of lambda in our template in cell B3 to be 0.05. And the average then gets updated to 20 minutes. And Tejas, what does that come out to be from our template? We get the probability of X less than or equal to 15 to be equal to 0. 632. Correct. That means there is a 63 percent chance that his team will finish the oil change in less than 20 minutes. It looks like Rohit is going to give a lot of discounts.

Next, Rohit wants to know what is the probability that an oil change will take anywhere between 15 and 30 minutes. We are modeling the oil change time X as an exponential random variable with lambda equal to 0. 05. Ashwini, help us out with this one. Yes, now we want the probability that capital X lies between values of 15 and 30.

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Yes, and Tejas, what do we get from our template? When we put those values of capital A equals to 15 and capital B equals to 30 in the template, we get the probability to be 0. 249. Yes, that is correct. So there is a 25 percent chance, roughly, that the oil change will be completed between 15 to 30 minutes.

Next, Rohit wants to know what is the probability that his team will do more than 15 oil changes during a four hour morning shift. Notice that in this case, we are now interested in the number of oil changes rather than the time taken to do an oil change. That is, we have switched from the continuous random variable X that corresponds to the time for an oil change to the associated discrete random variable capital Y that corresponds to the number of oil changes being done in a given time. Tejas, what would be the distribution of capital Y, the number of oil changes done in a given time? Since the time for an oil change is exponential, the discrete random variable corresponding to the number of oil changes would be a random variable.

Yes, and Ashwini, what would be the average or the capital lambda corresponding to the number of average oil changes done during a four hour morning shift? Since it takes 20 minutes on an average for an oil change, in four hours we would expect to do 12 oil changes. So capital Lambda would be equals to 12. Excellent. Notice how we use the average of 20 minutes for the exponential and the time period of four hours to compute the average for the Poisson, capital Lambda. Capital lambda is equal to small lambda times t. Now, we want to know the probability of capital Y being greater or equal to 15 for a Poisson random variable with capital lambda equal to 12.

Let's go to our Poisson template to compute this probability. We update the value of capital lambda to be equal to 12. And then we read out the probability of y greater or equal to 15 from cell E21. It comes out to be 0.228.

Next, Rohit wants to know what is the probability that his team will do more than 30 oil changes during an entire eight hour day. Notice, here again, we are counting the number of oil changes. However, the time period of interest has shifted from a four hour morning shift to an eight hour day. That would imply that we have a different version of the Poisson random variable. We are now interested in a Poisson random variable, capital Y, corresponding to the number of oil in an 8 hour day.

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The average number of oil changes in an 8 hour day would be equal to 3 times 8, which would come out to be 24. In other words, the capital lambda would be equal to 24. So, Ashwini, can you now find out the probability of capital Y being greater or equal to 30 for a Poisson random variable that has an average of 24.

Yes, first I update the value of capital lambda in the template to 24 and read the probability of capital Y being greater or equal to 30 from cell E36 to be 0. 132. Both these do not look very promising. I should tell Rohit bhai, he is making a mistake. You are right, his scheme is too aggressive.

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The Normal Random Variable and its Distribution

Now that you have a good feel for the uniform and the exponential random variable. Let's look at perhaps arguably the most important continuous random variable, the normal random variable, and its distribution. The most commonly used probability distribution for describing a continuous random variable is perhaps the normal probability distribution.

The normal distribution has been used in a variety of applications. For example, the normal distribution is used to model the height or weight of a sample of people that may be surveyed. It is also used to model test scores of students. It finds several applications in scientific measurements and experiments.

The normal distribution is also used to measure the amount of rainfall in a particular location and several other weather related applications. It is also most widely used in statistical inference, which is the main reason we are discussing it now. We will use the properties of the normal distribution a lot, especially when we try to make inferences about a population using the results obtained from a sample that we have drawn from that particular population.

So let's begin by looking at the normal distribution and its properties. In probability theory, the normal distribution is the most commonly used continuous probability distribution. The credit for research related to normal distribution is often given to two famous researchers. The first is Johann Carl Friedrich Gauss. He was a German mathematician and physicist who made significant contributions to many fields in mathematics and science. So sometimes the normal distribution is also called as the Gaussian distribution. The second is Abraham de Moivre. He was a French mathematician and is well known for his DeMoivre's formula and for his work on the normal distribution in his famous publication, The doctrine of chances.

However, the normal distribution is called so and not usually by the names of these researchers because it models the randomness in many normally occurring phenomenon. It is also often referred to as the bell shaped curve because of the shape of its PDF and the symmetry about its mean. The form and shape of the normal distribution is depicted here.

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It is the famous bell shaped curve. The probability density function that defines this bell shaped curve for the normal distribution is given as follows. The small f of x is equal to 1 over sigma times the square root of 2 times pi multiplied by e raised to minus of x minus mu square divided by 2 times sigma square.

For the beautiful bell shape, that is a messy probability density function. Yes, Ashwini, that's right. And imagine integrating that to get cumulative probabilities and tail probabilities. Are we going to have to do that? Fortunately, no. Because of its numerous applications, the calculations related to the probability density function have been well studied and analyzed.

There are several tables that record these probabilities that avoid the need for us to do the integration of the PDF for most applications. Let's first look at some of the properties of this probability density function. First, the entire family of normal distributions is differentiated by two parameters, the mean mu and the standard deviation sigma.

Second, the highest point on the normal curve is at the mean, which also happens to be the median as well as the mode of the distribution. Third, the mean of the normal distribution can take any numerical value, negative, zero or positive. Technically, the value of the normal random variable itself can range from minus infinity all the way to plus infinity.

This is unlike the exponential random variable that we just saw. Do you remember the range of values that it can take? Yes, it was defined only for values of zero or greater. Precisely. Fourth, the normal distribution is symmetric with the shape of the normal curve to the left of the mean. being a mirror image of the shape of the normal curve to the right of the mean.

Also, the tails of the normal curve, both the left tail and the right tail, extend to infinity in both directions. and theoretically they never touch the horizontal axis. Also, because it is symmetric, the normal distribution is not skewed and its skewness measure is equal to zero. Fifth, the standard deviation determines how flat and wide the normal curve can be.

Larger values of standard deviations result in a wider and flatter bell shaped curve, showing more variability in the data. A narrower normal distribution leads to a skinnier and taller bell shaped curve, indicating less variability in the data. The probabilities of the normal random

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variable are given by the area under the normal curve, just like with any other probability, density function.

Also the total area under the curve for a normal distribution would like any other PDF be equal to one now, Ashwini? What would be the area under the curve to the left of the mean? We'll have to integrate. Not really, because the distribution is symmetric. The area under the curve to the left of the mean would be exactly equal to 0. 5. Oh, yes. And the area under the curve to the right of the mean would be exactly equal to 0. 5. Yes. Another common and popular property of the normal random variable is related to some of the percentage values that are commonly used in practice related to the spread of the data. For example, it is known that 68. 3 percent of the values of a normal random variable are within plus or minus one standard deviation of its mean. What this implies is that if X is a normal random variable with mean mu and standard deviation sigma, then the probability that this random variable capital X lies in the range mu plus sigma and mu minus sigma is going to be equal to 0. 683. Similarly, 95. 4 percent of the values of a normal random variable are within plus or minus two standard deviations of its mean. That is, if x is a normal random variable with mean mu and standard deviation sigma, then the probability that the random variable capital X lies in the range mu plus 2 times sigma and mu minus 2 times sigma is going to be equal to 0. 954. And finally, 99. 7 percent of the values of a normal random variable lies within plus or minus 3 standard deviations of the mean. That is, if X is a normal random variable with mean mu and standard deviation sigma, then the probability that the random variable capital X lies in the range mu plus 3 times sigma and mu minus 3 times sigma is going to be equal to 0. 997.

A random variable that has a normal distribution with a mean 0 and standard deviation of 1 is said to have the standard normal The letter capital Z is commonly used to designate this particular normal random variable instead of the usual letter capital X. Here you see a graph of the standard normal distribution.

It has the same general bell shape appearance as any other normal distribution, but with special properties that its mean mu is equal to 0 and its standard deviation sigma is equal to 1. Because the mean is equal to 0 and the standard deviation is equal to 1, the formula for the standard

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normals probability density function is simpler than the pdf of the regular normal distribution that we just saw.

We can write the pdf of the standard normal distribution denoted by small f of z to be equal to 1 divided by the square root of 2 pi multiplied by e raised to minus z squared divided by 2. In some books, you will find that they use the Greek letters of small phi and capital phi instead of small f and capital F to denote the pdf and the cdf of the standard normal random variable.

Why this change of notation? Yes, is there a reason? I am not aware of a specific reason, but the standard normal is a special random variable, and it does deserve special treatment. Besides, when you see small phi and capital phi, we immediately know that we are talking about the standard normal. Anyway, as with all continuous random variables, the probability calculations for any normal distribution is made by computing the areas under the graph of the probability density function. For the standard normal distribution, area under the normal curve has been computed so that Ashwini and all of you do not have to do the integrations. Yes, I am glad I don't have to do the integrations. And these probabilities are available in various tables. You can see a snapshot or a picture of these tables right here.

And like with other random variables, we are usually interested in three types of probabilities. The first is the probability that a standard normal random variable Z takes a value less than or equal to a given value. What do we call this? The cumulative distribution denoted by capital F or capital Phi in this case.

Yes. And the second type of probability that we are interested in is the probability that the standard normal random variable Z takes values between two given values, say A and B. Yes, we did this with the other random variables like uniform and exponential. That is true. And the third is the probability that the standard normal random variable Z will be greater than or equal to a given value.

What did we call this? The tail probability, of course, denoted by f bar or phi bar in this case. Yes, and we will use an Excel template to calculate the probabilities corresponding to the normal distribution. Let's see the details of the template next before we use it to solve some

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examples. We will set up the template for any normal distribution, and then the computations for the standard normal will just be a special case of what we can do.

Consider a normal distribution with a specified mean mu and a standard deviation sigma, like we did for the other distributions we have seen so far. We will use some inbuilt functions in Excel. In particular, we will use the norm.dist function to compute the cumulative probabilities of a normal distribution.

The norm. dist function in Excel has four parameters. The first parameter is small x, which corresponds to the value of the normal random variable for which we want to compute the cumulative probability. The second parameter is mu, or the mean of the normal random variable that we are analyzing. The third parameter is sigma, the standard deviation corresponding to this normal distribution.

The fourth parameter is a toggle, which takes a value of true or false. When the toggle is set to true, the function provides us with the cumulative probability corresponding to small x and the other parameters that we have specified. When the toggle is set to false, it provides the probability density function value corresponding to small x and the other parameters specified.

Excel also has a function called norm dot in, which uses an inverse computation. We use this function when we want to find the value of small x for a given value of the cumulative probability. The norm in function has three parameters. The first parameter corresponds to the probability for which we want to find the inverse. Of course, it has to be a value between 0 and 1. The second parameter corresponds to mu, the mean of the normal distribution that we are analyzing. And the third parameter is sigma, corresponding to the standard deviation of the normal distribution that we are analyzing. Let's see how these are incorporated in our template.

Here in cell B3, I have entered the value of the mean of the normal distribution, mu. In this case, it is equal to 10. And then in cell C3, I have entered the standard deviation, sigma. In this case, it is equal to 3. Next, there are two sections below these inputs. The first section is used for probability computations, and the other section can be used for computing the inverse calculations.

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In both sections, you can enter values in the green cells and see the outputs in the white cells. In the first section, by entering a value of small x, you can compute the cumulative distribution. The probability of x being less than or equal to some value small x, or the tail probabilities f bar of small x, which is the probability that x is greater than or equal to small x.

And we can also compute the probability that small x takes values between some value of a and some value of b. When I enter the value of 13, you will notice that we get the probability of 0. 841 in this case for the probability that x takes a value less than or equal to 13. When I enter a value of 13 in the tail probability section, we get the tail probability of probability x greater or equal to to be equal to 0. 159.

And finally, if I enter A equal to 7 and B equal to 13, then I get the probability of X lying between 7 and 13, both inclusive. To be equal to 0.683? Yes. Since we're going one standard division on either side, we should get 0.683. That's a good observation. Ashwini in the section below used for inverse calculations.

You can find the value of small x if you are given the value of either the cumulative probability. Probability of x being less than or equal to some value small x, or the tail probabilities f bar for a small x. If I enter the probabilities that we just got for the cumulative probability of 0. 841, I get the value of x to be equal to 13, as I should. And if I enter the value of 0. 683, I get a value of a to be equal to 7, and b equal to 13. This is very helpful. Yes, and on the right hand side, we have a table in columns P, Q, and R that computes the PDF and the CDF for the standard normal distribution for various values of small z. You can also see a related plot of the PDF.

Okay, let's get some practice in calculating probabilities for the normal distribution. using these templates. Here is the first one. Given that Z is a standard normal random variable, let's compute the following probabilities. The probability that z is less than or equal to minus 1, the probability that z is greater or equal to minus 1, the probability that z is greater or equal to minus 1. 5, the probability that z is greater or equal to minus 2. 5, and finally, the probability that z lies greater than minus 3, but is less than or equal to 0. Let's get started so that we can use the template that we just created. We update the mean to be equal to zero and the standard deviation to be equal to one, to correspond to the standard normal, random, variable Z.

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Then the first question asks us to find out the probability of Z being less than or equal to minus one. We put the value of minus one in the cumulative probability distribution section. We find out the probability to be equal to 0.159. Next, we want to compute the probability of Z being greater or equal to minus one.

We use the tail probability section and we enter the value of minus one and read out the probability to be equal to 0.841. Aren't we double counting the probability of Z being exactly equal to minus one in both of these cases pages? Think through the question you just asked me and tell me whether we are double counting it.

What is the probability of z being exactly equal to minus one? Oh, I know. Wouldn't that be equals to zero? So we are not double counting. Yes, it would be equal to zero, but why? Because for a continuous random variable, the probability of the random variable taking any specific value is always equal to zero.

Yes, exactly. For continuous random variables, we always define probability only for intervals. And the probability of a random variable taking a single value is always equal to zero. Alright, next we want the tail probability corresponding to minus 1. 5. So we enter minus 1. 5 in the template, in the tail probability section, and we read out the probability to be equal to 0. 933. Then we need to compute the tail probability corresponding to minus 2. 5. We enter that into the template and we compute the tail probability to be equal to 0. 994. Finally, we want the probability that z takes a value between 0 and minus 3. We enter these values in the relevant section and we get the probability to be equal to 0. 499. Good.

Let's do one more example. Given that Z is a standard normal random variable, Let's compute the following probabilities. The probability that z lies between 0 and 0. 8. Ashwini, what is it? We have a equals to 0 and b equals to 0. 8, giving the probability to be 0. 288. Correct. And Tejas, what would be the probability that Z lies between minus 1. 5 and 0? Now, A is equal to minus 1. 5 and B is equal to 0. The probability is 0. 433. Good. And Ashwini, can you give me the probability that Z is greater or equal to 0. 4 and the probability that Z is greater or equal to minus 0. 2. We use tail probabilities now. We get probability of z greater than or equals to 0. 4 equals to 0. 345 and the probability that z is greater than or equals to minus 0. 2 is 0. 579. Correct. And Tejas, can you compute the probability that z is less than or equal to 1. 2 and the

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probability that z is less than or equal to minus of 0. 7? We need the CDF now. We get probability that probability of z lesser than equals to 1. 2 equals to 0. 885. And the probability of Z lesser than equals to minus 0.7 equals to 0.242. Well done both of you. Now let's do an inverse calculation.

Given that Z is a standard, normal, random variable, let's find the range of Z that satisfies the following conditions. The area to the left of Z is 0.21. Now, Ashwini, how would you calculate the range of z that satisfies this value? We need to use inverse and in the CDF section, we put 0. 25 and we get that for z less than minus 0.806. The CDF would be 0. 21. Excellent. How about this? For what values of z do we have the area between minus z and plus z to be equal to 0. 9? We use the inverse section again and we put the value of 0. 9 and we get the values of minus 1. 64 to plus 1. 64. Correct. And Ashwini, can you give us the range of z so that the area between minus z and plus z is 0. 2? It is similar to what Tejas just did. I get z should lie between minus 0. 25 and plus 0. 25. Okay. Tajus how about the range of z so that the area to the left of z is 0.955? We use the CDF section in the inverse part. I get that it holds for Zed less than 2.58, is that correct? Yes, it is and Ashwini one final.

Can you give us the range of z so that the area to the right of z is 0. 692? We now use the tail probability part in the inverse section and get that for z more than minus 0. 502, the tail probability is 0. 692. Wonderful.

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Standard Normal Distribution

The concept of critical values is widely used in statistical inference. So next, we are going to see this concept in relation to the standard normal distribution. We know that the standard normal distribution is a symmetric bell shaped distribution. It is centered at 0 with a standard deviation of 1.

For this standard normal random variable Z, we define a critical value known as Z alpha by 2. Corresponding to a given value of alpha, we define two critical values of Z known as Z alpha by two and minus Z alpha by two. The interpretation is as follows, the area or the probability to the right of Z alpha by two, that is the tail probability corresponding to Z alpha by two is equal to alpha by two.

Similarly, the value of minus Z alpha by two is that value of Z, such that accumulative probability corresponding to minus z alpha by 2 is equal to alpha by 2. Therefore, the probability that z lies between minus 2 and plus z alpha by 2 is going to be equal to 1 minus alpha. Now, this is a generalization of a concept that we have already seen.

For example, we know that for z equal to plus or minus 1 we have 68 percent of the values lying in between. This implies that for 1 minus alpha equal to 0. 68, we would have alpha by 2 equal to 0. 16. Then the z critical corresponding to alpha by 2 of 0. 16 would be plus 1 and minus 1. Let us now compute the critical values of z alpha by 2 corresponding to the various values of alpha or 1 minus alpha.

When 1 minus alpha is equal to 0. 8, we have alpha by 2 equal to 0. 1. And then from our inverse probability calculations we can get the critical value of Z Alpha by two or Z of 0.1 to be equal to 1.28. When one minus alpha is equal to 0.9. We have alpha by two equal to 0.05 from our inverse probability calculations we can get the critical value of Z alpha by two or Z of 0.05 to be equal to 1.64. When one minus alpha is equal to 0.95. We have alpha by two equal to 0.025, and again from our inverse probability calculations. We can get the critical value of Z alpha by two or Z of 0.025 to be equal to 1.96. When one minus alpha is equal to 0.99, we have alpha by two is equal to 0.005, and again from our inverse probability calculations.

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We can get the critical value of Z alpha by two or Z of 0.005 to be equal to 2.58. These critical values will be used quite a bit when we consider the applications of the normal distribution in statistical inference. In fact, we will do similar calculations on other distributions that are also used in statistical inference.

The examples and calculations that we just did illustrate how the template or the tables that provide the cumulative probabilities for the standard normal distribution can be used to find the probabilities associated with the value of the standard normal random variable z.

As we saw in the examples. We are usually interested in two types of questions. The first type of question gives us the value or values of z and then asks us to find the corresponding areas or probabilities. The second type of question is usually the inverse calculation, where you are given the area or the probability and then we are asked to find the corresponding value or values of z.

Hopefully, you're all comfortable answering both types of questions. One tip that usually works is to sketch a graph of the standard normal probability distribution and shade the appropriate area based on the question being asked. This helps to visualize the situation and usually makes it easier to find the correct answer.

Let's now move on from the standard normal distribution. The reason for discussing the standard normal distribution so extensively is that the probabilities for all normal distributions regardless of the mean mu and the standard deviation sigma can be computed using the standard normal distribution.

All normal distributions can be analyzed using the standard normal? Yes, exactly. If you have a random variable x that has a normal distribution with some mean mu and some standard deviation sigma. We can answer probability related questions about this random variable X by first converting it into a standard normal distribution.

Then we can use the standard normal template to find the appropriate values of Z or the desired probabilities and then translate it back to answer the question about X. It's like converting

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temperature from Celsius to Fahrenheit, our currency conversion from dollars to a rupees and so on. Yes, we convert the problem into a problem related to the standard normal.

We find the answer and then convert it back. The formula used to convert a normal random variable X with some mean new and some standard deviation. Sigma to the standard normal, random variable Z is as follows. We write z is equal to x minus mu divided by sigma. A value of x equal to its mean mu will of course result in a z equal to mu minus mu divided by sigma which is equal to zero.

That is, when the value of x is equal to its mean, the corresponding value of Z is equal to zero. Now, suppose that X is one standard deviation above the mean, that is X is equal to new plus sigma. Then the equation for Z tells us that the corresponding value of Z is going to be equal to mu plus sigma minus mu divided by sigma, which would come out to be equal to one.

So the value of x when it is one standard deviation above the mean corresponds to z equal to one. Notice that if x was mu minus sigma, that is below the mean by one standard deviation, then the corresponding value of z would be equal to minus 1. That means that we can interpret z as a number of standard deviations random variable X is away from its mean, either above or below.

And the value of Z can be positive when X is above its mean, and the value of Z can be negative when X is below its mean. This conversion helps us compute the probabilities for any X that is above that has a normal distribution using the calculations for the corresponding standard normal z. Suppose we have a normal distribution with mean mu equal to 10 and sigma, the standard deviation, equal to 2.

And we want to find the probability that this random variable capital X lies between 10 and 14. Using the conversion equation, we see that when x is equal to 10, Z is equal to 10 minus 10 divided by 2, that is equal to 0. And when X is equal to 14, Z is equal to 14 minus 10 divided by 2, which is equal to 2.

Thus, the probability of X being between 10 and 14 is equivalent to the probability that the standard normal random variable Z takes values between 0 and 2. Now, we can go to our

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template and compute this probability. And we get that equal to 0. 4772. We could use our template directly for x using a mean of 10 and standard deviation of 2, right?

Why convert it to z at all? Yes, Ashwini, we could have done that too with our template and we should get the same results. So, given our template, it is not necessary to convert any normal random variable X to the corresponding Z, but it is always helpful to know that there is an equivalence here.

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Examples of Normal Distribution

Munaf Khan is a stock broker in Baruch and he's been tracking large cap stock funds for the last three years. He has observed that it provides a mean return of 12 percent and a standard deviation of 3%. He considers an investment to be excellent if the investment returns more than 10 percent in a year.

He considers an investment to be a moderate investment. if it returns between 5 and 10 percent in a year. And he considers an investment to be an unsatisfactory investment if it returns less than 5 percent in a year. He wants to know what is the probability that his investment falls into one of these three categories.

He also wants to know what is the 90th percentile of the returns he can expect. Ashwini and Tejas, let's analyze this. First, we can assume that the annual percentage returns are normally distributed with a mean of 12 percent and a standard deviation of 3 percent. Next, we go to our normal distribution template and we update the mean in cell B3 to be equal to 12 and the standard deviation in cell C3 to be equal to 3 percent.

Then, we want to know the probability that the investment would be unsatisfactory. That means it has to give a return of less than 5%. We enter 5 percent in our cumulative distribution section, and we get our corresponding probability to be 0. 01. Next, we look at the probability of it being an excellent investment, which is giving us more than 10 percent return. Tejas, how do we do this? We need the tail probability corresponding to x equal to 10. Correct, Tejas. So we put 10 in the tail probability section and we get the corresponding probability to be equal to 0. 748. In other words, there is close to a 75 percent chance that this would be an excellent investment.

Third, we want to look at the probability that it is a moderate investment. That is, it gives us returns between 5 and 10 percent. Ashwini, how will we do this? Shouldn't we just set A equals to 5 and B equals to 10 in the next section of the template and find the corresponding probability? Yes, that is right, Ashwini. And if I did that, we get the probability to be equal to 0. 243. So, there is a 24 percent chance that this is a moderate investment. Finally, we want to have the 90th percentile of the returns that Munaf can expect. Ashwini, how will we get this?

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We need to use the inverse probability calculations. That's right. We can put 0. 9 in the inverse section, both corresponding to the cumulative probability and corresponding to the tail probabilities.

Putting it in the cumulative probability tells us that there is a 90 percent chance that this investment will give us a return of less than 15. 8 percent in a year. Yes, and putting 0. 9 In the tail probability section tells us that there is a 90 percent chance that he will get more than 8. 1 percentage returns in a given year. The final section of the inverse probability tells us that there is a 90 percent chance that this investment will give us a return between 7 and 17%. Very good.

Let's now visit Bhavani at the Sharada Hospital in Gandhinagar. Bhavani Kapadia is an ophthalmologist at the Sharada Eye Hospital in Gandhinagar. She conducts a study of screen times among school kids. She finds out that the screen time per day has a normal distribution with a mean equal to 8. 4 hours and a standard deviation equal to 2. 5 hours. She is visiting the local school and wants to present some facts to the kids and parents. She wants to know what is the probability that a kid has screen time between 6 to 12 hours in a day.

How many hours of screen time must a kid have in order to be in the top 20 percent of the kids? Third, she wants to know how many hours of screen time must a kid have in order to be in the bottom 20 percent of kids. And finally, she wants to know what is the probability that a kid has more than four hours of screen time per day.

Tejas and Ashwini, let's help Bhavani. Tejas. Can you get us started with the first one, which is Bhavani wants to know what is the probability that a kid has more than four hours of screen time in a day. We can use our template and set the mean equal to 8. 4 and the standard deviation equal to 2. 5. We go to the probability section and calculate the tail probability for x equal to 4 and we get the probability to be 0. 96.

Next, Bhavani wants to know what is the probability that a kid has screen time between 6 and 12 hours per day. We enter 6 and 12 in the probability calculation section for A and B. And if you did that, you will get the probability to be equal to 0. 757. That is, 76 percent of the kids have between 6 to 12 hours of screen time per day.

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Next, Ashwini let's calculate the hours of screen time a kid must have in order to be in the top 20%. To be in the top 20%, in terms of screen time, 80 percent of the kids should have screen time less than our desired value. Yes, that is correct. We need to use the inverse calculation here. The bottom section. To find the top 20%, we need the cumulative probability to be equal to 0. 8. We put in the desired probability to be 0. 8, and we get the value to be 10. 5 hours. That means 80 percent of the kids have 10 and a half hours of screen time or less. Therefore, a kid having more than 10 and a half hours of screen time would have to be in the top 20%.

Bhavani also wants to know, how many hours of screen time should a kid have to be in the bottom 20%? Tejas, can you help with this one? This is also an inverse calculation. And now, we'll look at the tail probability. Yes, and we put the tail probability to be equal to 0. 8. And we read out the probability to be equal to 6. 3 hours. That means that 80 percent of the kids have more than 6. 3 hours of screen time per day. Just for curiosity, if you put 0. 8 in the interval range, you get that 80 percent of the kids have between 5. 2 hours and 11. 6 hours of screen time every day. Tejas and Ashwini, what do you think about screen times? Do you worry about that? Yes, between courses, job and phone calls, I think I have a lot of screen time. Me too, but it is all digital these days. I don't think we have much of an option. Yes, that's true. Tough choices, aren't they?

Now, let's go to the historic city of Vadodara and visit Hirav at the Maharaja Sayajirao University that was established in 1881. Did you know that Vinoba Bhave & Dadasaheb Phalke were alumni of this university? No, I did not. Hirav Joshi teaches physics at the Maharaja Sayajirao University in Vadodara. It is one of the oldest physics departments in the country. But Hirav has a reputation among students for setting tough exams. His exams are usually open book, open notes with an open time limit. Oh, I never do well in these type of exams. Yes, open book means more difficult exams.

Hirav has recorded the time taken for students to complete his exam. He has observed that it has a normal distribution with a mean of 150 minutes and a standard deviation of 20 minutes. There are 60 students taking his class. They are interested in the answers to some questions. They want to know how many students are likely to complete the exam in 2 hours or less. They

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would like to know how many students will take between two and three hours to complete the exam.

Finally, how many students are expected to take three hours or more to complete the exam. Ashwini, let's first update our template and then answer these questions. We set the mean to be 150 and the standard deviation to be 20 minutes. Ashwini, how many students are likely to complete the exam in 2 hours or less?

We need cumulative probability corresponding to 120 minutes, that is 0. 067. Yes, that would imply 0. 067 times 60, which is roughly 4 students. Everyone is going to be writing a lot. Tejas, how many students will take between two to three hours to complete the exam? Ashwini has already updated the template. We set A to be two and B to be three hours in the probability section and we get the probability to be close to zero. That does not make sense. Yes, it does not. Do you know what you did wrong? You need to enter it in minutes. 2 hours is 120 minutes and 3 hours is 180 minutes. Yes, you need to be careful about the units.

If you enter A equal to 120 and B equal to 180, you get the probability to be equal to 0. 866. which corresponds to roughly 52 students out of the 60.

It is always a good idea to do a little sanity check when you do these probability calculations. Finally, how many students are expected to take 3 hours or more to complete the exam? We put 180 minutes in the tail probability section and we get the probability to 0. 067 or 4 students. We got 4 students for the first one too.

Yes, and do you know why? The first question had 120 minutes, which was one and a half standard deviations below the mean of 150 minutes. This question had 180 minutes, which is also one and a half standard deviations, but above the mean. Oh, by symmetry of the normal distribution, the probabilities would be the same.

Yes, exactly. Now, let's visit a sugar factory in Navsari, where Naman Shah is waiting for us. Naman Shah works at a sugar mill in Navsari. The sugar packets produced are supposed to have one kg weight. Naman has a choice of two filling machines that he can use at his factory. One filling machine is known to fill the packets with sugar quantity that is normally distributed

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with a mean of 1. 01 kg and a standard deviation of 0. 02 kg. The second filling machine is known to fill packets with a sugar quantity that is also normally distributed, but with a mean of 1. 03 kgs and a standard deviations of 0. 04. Both of these machines are not perfect, and Naman wants our help to find out which one of these machines might be a better choice.

His first question is What proportion of packets filled by these machines are likely to be underweight? That is weighing less than one kg. Second, he wants to know what percentage of packets filled by these machines are are going to have a weight between 1 and 1. 05 kg. He also thinks that 90 percent of the packets filled by either of these machines are going to have a quantity that lies between 0. 99 and 1. 01 kg. Is his claim true?

Let's find out. Ashwini, Let's work on these questions for the first filling machine. We know that for the first filling machine, the weight filled is normally distributed with a mean of 1. 01 and 0. 02 standard deviation. Yes. So now what is the probability that this machine is going to fill packets that are weighing below the desired 1 kg?

We update our template to have a mean of 1. 01 and a standard deviation of 0. 02. Then we compute the cumulative probability of having packet being less than 1 kg in weight to be 0. 309. Next, we want to know what percentage of packets lie between 1 and 1. 05 kg. We put these as A and B in our template, and then we can read out the probability for this machine to be equal to 0. 669.

That is, roughly 67 percent of the packets are going to lie between 1 and 1. 05 kgs. Finally, let's look at his claim about 90 percent of the packets being in a given range. This is an inverse calculation. So, if you put 0. 9 in the inverse probability, we get the range to be equal to between 0. 97 and 1. 04.

So, in this case, Naman's claim is not really true. But how different is this for the other filling machine? Tejas, let's find that out. For the second filling machine, the weight filled is normally distributed with a mean 1. 03 and 0. 04 standard deviation. Yes, and so what is the probability that this machine is going to fill packets that are weighing below the desired 1 kg?

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Don't forget to update the template and have a mean of 1. 03 and a standard deviation of 0. 04. We compute the cumulative probability as 0. 227. That is, there is a 23 percent chance of having packet less than 1 kg in weight. It is better than machine one. Next, we want to know what percentage of packets lie between 1 and 1. 05 kgs. We put these as A and B in our template, and then we can read out the probability for this machine to be 0. 465.

Seems better again? Yes, so roughly 47 percent of the packets are going to lie between 1 and 1. 05 kgs. Finally, let's look at his claim about the 90 percent of the packets being in a given range for this machine. Again, this is an inverse calculation. So you put 0. 9 in the inverse probability section and we get the range to be equal to between 0. 96 and 1. 09. So, in this case, Naman's claim is not true again, and the second machine is worse on this count. Yes, it is indeed a tough choice for Naman.

Let's now go to Mr. Kapadia in Ankleshwar. Vibhor Kapadia works at a glass factory in Ankleshwar in Gujarat. His factory makes fine glass that is used in a lot of architectural applications. He needs help with a quality problem. He has a machine that makes glass sheets that are supposed to have a thickness of 3 millimeters. Although the machine is correctly calibrated, the thickness of the sheets coming out vary. The thickness of the glass sheets coming out has been found to be normally distributed with a mean of 3 millimeters and a standard deviation of 0. 12 millimeters. He wants to know what is the probability that the glass sheets that come out of his machine will have a thickness less than 2. 9 millimeters. He wants to know what is the probability that the glass sheets will have a thickness of more than 3. 2 millimeters. He also wants to know what is the thickness range within which 95 percent of his glass sheets are going to lie.

First, let's update the template to reflect a normal distribution with a mean of 3 millimeters and a standard deviation of 0. 12 millimeters. Now, we want to know the probability that the thickness is less than 2. 9 millimeters. We go to the cumulative probability section and enter 2. 9 and read off the probability to be equal to 0. 202. That is 20 percent of his glass sheets are going to be less than 2. 9 millimeters thick. Next he wants to know the probability that the glass sheets have a thickness of more than 3. 2 millimeters. Now we are interested in tail

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probabilities. We enter 3. 2 millimeters in the tail probability section and read off the probability to be equal to 0. 048, or roughly 5%.

Finally, he wants a 95 percent guarantee for the thickness range produced by his machine. So this is an inverse calculation. So we enter 0. 95 in the inverse probability section and we read out the corresponding values of a and b.

The value of a comes out to be 2. 76 and b comes out to be equal to 3. 24 millimeters. So, we can say that 95 percent of his glass sheets made by his machine are going to lie between 2. 76 millimeters and 3. 24 millimeters.

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Normal Approximation to the Binomial Distribution

A very useful property of the normal distribution is that it provides good approximations to the probability values of certain other distributions. Especially in cases where the cumulative distribution function is a rather complicated distribution. For all the complexity we saw related to the pdf of the normal distribution, I am sure you agree that computing the probabilities corresponding to the normal was fairly easy.

Yes, calculating the probabilities using the template was quite easy and fun. Yes. And we want to take advantage of this. One of the most common examples of doing this is when a normal distribution is used to approximate the binomial distribution. Tejas, do you recall the probability mass function of the binomial distribution? That is, the function that gives us the probability that a binomial random variable capital X takes values equal to small x. Let me refer to my notes here. Found it. I think it was n choose x times p to the power x multiplied by q to the power of n minus x, where p and q are the success and failure probabilities respectively.

Yes, and the term n choose x involves factorial computations, which can be challenging, especially if n is very large, in thousands, for instance. In such cases, we can approximate these binomial probabilities very well by the probabilities corresponding to a normal distribution. We will see some examples of this in a bit.

But you know, the use of the normal approximations is not just limited to the binomial. In fact, in some cases, the normal is used to approximate the Poisson distribution too. That is interesting. The Poisson has an e to the power minus lambda term in its PDF. I guess that can be hard to compute also. Yes, and that is one of the motivations.

However, a little later, we will see a very important theorem known as the Central Limit Theorem or CLT. This theorem argues that the normal distribution provides a good approximation for the distribution of an average of a set of identically distributed random variables. irrespective of the distribution of the individual random variables.

But don't worry about that now. We will get to it a little later. Let's go back to see how the normal distribution approximates the binomial distribution. Wait a minute. Let me make sure

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I'm understanding this correctly. The binomial distribution is a discrete distribution and the normal distribution is a continuous distribution.

And are you saying that we are going to approximate the discrete distribution by a continuous distribution? Yes, Ashwini, you heard me correctly. We are going to approximate a discrete probability distribution by a continuous probability distribution. Seems intriguing and perhaps ambitious, isn't it? Yes, you have made me curious.

But first, let's look at a plot of both of these distributions. If you look at the probability mass function of the binomial distribution, you see that it is by and large bell shaped and similar to the probability density function of the normal distribution. And that is really the motivation. It turns out that the normal distribution is very good at providing approximations to the probability values of a binomial distribution, especially when the parameters of the binomials are reasonably large which is really the case where we need help with these approximations. And when the success probability p is not very close to 0 or 1, particularly when n times p is greater or equal to 5 and n times q is also greater or equal to 5. Tejas, since you have your notes about the binomial in front of you, what is the expectation and the variance of a binomial distribution with parameters n and p?

I see here that the expectation of n times p and the variance is n times p times q. Correct. So the first step in approximating a binomial distribution with a normal distribution is to match the mean and the variance of the binomial and the normal distribution. So, we consider a normal distribution with mean equal to n times p and variance equal to n times p times q.

Let's do this using an example. Suppose we toss an unbiased coin 16 times. It has success and failure probabilities of p equal to 0.5. Then the expectation is, expectation is 8 and variance is 4. We can see the probability mass function of the distribution of the binomial random variable capital X right here along with the probability density function of the normal random variable capital Y with the same mean of 8 and variance of 4.

Even though the distribution of the random variable X is discrete and the random variable Y is continuous, the shapes and the respective probability mass function and probability density

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function are quite similar. Yes, Tejas, the bars of the probability mass function of capital X are not exactly the same height as those of the probability density function of capital Y.

But they are pretty close. So let us now verify how close they are by looking at a couple of examples. Let us use the binomial distribution to calculate the probability of the random variable X taking a value of 5 or less. Then we need to add the probabilities of X equal to 0, 1, 2, 3, 4, and 5.

Template time. I'm on it. Yes, indeed. So, we use our template for the binomial distribution. We update the value of n equal to 16 and p equal to 0. 5. Then, we look for the cumulative probability corresponding to the value of 5, and we get that equal to be 0. 105. Now, we compute an approximation for this probability, using the normal distribution of Y with a mean of 8 and a variance equal to 4. Template time again, this time normal distribution template. Yes, and let's input the values of the mean and variance. And then compute the probability that the normal random variable capital Y is less than or equal to 5. 5. You said 5. 5, not 5. But we need the probability of X less than or equals to 5. Yes. We have a slight correction of 0. 5 that is added to the value of 5. This is commonly known as the continuity correction.

And this correction is added in order to improve the accuracy of the approximation for the discrete binomial distribution by a continuous distribution or a continuous normal distribution. Ashwini, what did you get the probability from the normal distribution? I got 0. 1056. Then by making this correction, we would estimate the probability of the binomial random variable capital X taking a value of less than or equal to 5 to be approximately equal to the probability of a normal random variable taking a value less than or equal to 5. 5. We get that to be equal to 0. 1056. So you can see that the approximation provided by the normal distribution is pretty good.

Okay, let's look at another example to see how this works. Let's say we want to compute the probability that the binomial distribution takes a value between 8 and 11, both inclusive. That is, we want the probability that x is greater or equal to 8, but less than or equal to 11. So we will go back to our binomial template. And Tejas, can you read this out? Yes. So we would now add the probabilities of x equal to 8, 9, 10, and 11 from our template. And I get that equal to 0. 5598. Yes, I get the same value too.

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All right, so now let's see what we get using the normal approximation. Of course, for using the normal approximation in this case, we have to add a continuity correction again. The continuity correction in this case would be to look at the probability of y being greater or equal to 7. 5 and less than or equal to 11.5 Are we subtracting 0. 5 from 8? And also adding 0. 5 to 11? Yes, Tejas, that is correct. When we are looking at an interval, we subtract 0. 5 from the lower boundary of the interval, 8 in this case, and add 0. 5 to the upper boundary of the interval, 11 in this case. But if we needed only the CDF, like we just did earlier, we only increase the upper boundary by 0. 5, right? That is correct. When we are looking at only the cumulative probabilities, we increase the upper boundary by 0. 5. When we need the tail probability, we will subtract 0. 5 from the lower boundary.

And when we are looking at an interval, we do both. Alright, so now what do we get? We need to look for the probability that y takes a value between 7. 5 and 11. 5. From my normal template, using a value of a equals to 7. 5 and b equals to 11.5, I get the probability to be equal to 0. 5586. and the actual binomial probability was 0. 5598. So notice that the normal approximation seems to be remarkably good. Let me emphasize that the normal approximation to the binomial is better in terms of accuracy, as long as n times p is greater or equal to 5 and n times q is greater or equal to 5.

But I'm still wondering why we might use something like this at all, especially when we have a template and statistical software to compute them. Yes, as a user, these packages very well do the calculations for us. However, the computational challenge of computing these factorials still exists, whether it is a software package doing it, or whether you are doing it. And for all you know, the computer software may be computing these binomial probabilities using a normal approximation. Oh, I never thought of that. Just because it is not my problem does not mean that the problem does not exist. Yes, softwares could be using these approximations.

Let's look at one more example related to the normal approximation for the binomial. We are next visiting Kunal Poonawala in Junagadh. Tejas and Ashwini, have you ever been to Junagadh? No, I haven't. We were planning to go when we get a long weekend. If you get a chance, you must visit this town. It is famous for a lot of historical buildings, such as the Mahabat Magbara, the Upaharkot Fort, and of course, the Janagadh Museum.

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Our friend Kunal Poonawala works in a government office that is reconciling invoices obtained from all of these historical sites. He has trouble reconciling the errors in these invoices. Historically, it is known that roughly 10 percent of the invoices have some errors or the other. He has taken a sample of 100 invoices and he wants to compute the probability of exactly 12 invoices with errors in the sample of 100 invoices. That is, he wants to find the binomial probability of x equal to 12 successes in these 100 trials. Ashwini, do you see the irony here? Yes, we are denoting success in this case as an invoice with an error. Hardly something that you might consider is success. Yes, of course.

And Tejas, first give us the exact probability for this case from the binomial template. I set the value of n equals to 100. P equals to 0. 1, mean is 10, variance is 9, and probability of X equals to 12 is given by 0. 09. Good. Now, on to the normal approximation. The first step is to match the mean and the standard deviation. We look at a normal random variable capital Y with mean equal to 10 and standard deviation equal to 3. And what is the next step? Continuity correction. Probability of X equals to 12 is approximated by probability of Y lying between 11. 5 and 12. 5 inclusive. Yes, for a continuous probability distribution, the probability is always computed as the area under the probability density function, and always for an interval.

So we do not compute the probability for y equal to 12 from the PDF of the normal, as the probability of any single value of the random variable is going to be zero. So, to approximate the binomial probability corresponding to 12 successes. We compute the area under the corresponding normal curve between 11. 5 and 12. 5. We go to our normal template and update the mean to be equal to 10 and the standard deviation to be equal to 3. Then we let A to be equal to 11. 5 And B, to be equal to 12.5 and we read out the probability of Y being between 11.5 and 12.5 to be equal to 0.1052. And the exact probability, if you use the binomial distribution itself would be given by 0.09. So you see it is pretty close as an approximation.

Now suppose Kunal wanted the probability of 13 or fewer errors in his sample of 100 invoices. When we use the continuity correction factor, we will compute the y being less than or equals to 13. 5. Yes, and what do you get? We will get it to be equal to 0. 879. And the exact probability of having 13 or fewer invoices with errors computed using the binomial distribution itself would come out to be equal to 0.876.

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Notice that here too the normal approximation is a very good approximation of the true binomial probability. Now, before we move on, let me mention one other thing that I had briefly alluded to earlier. I had said that the normal distribution is sometimes also used to approximate the Poisson distribution.

And if you recall, for the Poisson distribution, the mean and the variances are both equal to capital lambda. Yes, that was one of the main characteristics of the Poisson distribution. Now, for the binomial, we have a mean equal to n times p and the variance equal to n times p times q. Now, if you consider a situation where the failure probability q is fairly small, then the variance of the binomial n times p times q could be very close to n times p, which is also its mean.

So in the cases where the failure probability q is small, the Poisson distribution can be approximated by the binomial. I can see where you are going with this. Yes, and now that we know that in some cases the binomial can be approximated by the normal, we can now connect the two and say that there are some cases where the Poisson distribution can be approximated by the binomial. Will we see an example? Not right now, but maybe later.

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Linear Combinations of Random Variables - I

Next, we will look at the distribution of linear combinations of normal random variables. Let us recollect what we had talked about when we talked about linear combinations of random variables in general, not necessarily for normal random variables. We had said that if X1, X2, and so on till Xn were a sequence of normal random variables, and a1, a2 and so on till an were some constants.

And if we defined a new random variable capital Y as a1 times x1 plus a2 times x2 all the way up to an times xn, then we could write the expectation of Y, E of y and the variance of y, var of y, using the expectations of the individual xis and the variance of the individual xis. In particular, the expectation of y, E of y, would be A1 times the expectation of x1 plus A2 times the expectation of x2, all the way up to An times the expectation of xn.

And if the individual xis are independent, Then we could write the variance of y, var of y, to be equal to a1 square times the variance of x1 plus a2 square times the variance of x2 and so on until an square times the variance of xn. However, if the xi's are not independent, that is, they are correlated, then the variance of y would be a1 square times the variance of x1 plus a2 square times the variance of x2 and so on until an square times the variance of xn plus a bunch of covariance terms corresponding to 2 times a i times a j times the covariance between x i and x j over all pairs i and j where i is not equal to j. Note that these results hold for any distribution that x i could take. What we are interested now is in a special case where the individual x i's are normally distributed.

So, of course, since we are talking of a special case, The results that we have said here for y and its expectation and variance would apply even if the individual xi's are normally distributed. The question is, can we say something more about y other than its expectation and variance? In general, it is hard to comment on the distribution of y if the individual xi's have some arbitrary or discrete continuous distributions.

However, if the individual Xi's have a normal distribution, then we can conclude that a linear combinations of these normal random variables given by Y not only has the expectation and variance given by the expressions that we just saw, but the distribution itself is also normal.

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Linear Combinations of Random Variables - II

An attractive property or feature of the normal distribution is that linear combinations of normal random variables are also normally distributed. As a simple case of linear combinations, consider Y to be a linear combination of a single normal random variable capital X with the form Y is equal to a times X plus b, where a and b are some constants.

If the random variable X has a normal distribution with a mean of mu and variance sigma squared, then Y also has a normal distribution. But with expectation of y being equal to a times the expectation of x plus b. And the variance of y would be a square times the variance of x. Now let's consider a special case.

Tejas and Ashwini compute this one. Let's say y is equal to a times x plus b, but a is equal to 1 over sigma and b is equal to minus mu divided by sigma. What do you get? The resulting y that I get is x minus mu divided by sigma. Isn't that z? Yes, it is. This particular linear function of x is essentially the standard normal random variable that we had discussed earlier.

Now, let's look at another simple case involving the summation of two independent random variables. That is Y is equal to X1 plus X2, where X1 is normal to with mean mu1 and variance equal to sigma1 square, and x2 is normal with mean mu2 and variance equal to sigma2 square. Let's assume that x1 and x2 are independent.

Then, y is equal to x1 plus x2 would also be normally distributed with mean equal to mu1 plus mu2. The variance of y would be equal to sigma1 squared plus sigma2 squared. However, if these two normal random variables x1 and x2 were not independent, then their sum would still be normally distributed. But the variance of Y would now be dependent on the covariance of these two random variables too.

We can now generalize this. If we had several independent normal random variables, Xi, for i equal to 1 through n, and each Xi with mean mu i and variance equal to sigma i squared, and Ai for i equal to 1 through n and b are some constants. Then, if we define a new random variable capital Y as a linear combination of these Xi's such that Y is equal to a1 times x1 plus a2 times x2 all the way up to an times xn plus b.

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Then, the expectation of Y would be equal to a1 times mu1 times mu 2 all the way up to a n times mu n plus b. And the variance of y, say denoted by sigma square, would be equal to a 1 times sigma 1 square plus a 2 times sigma 2 squared all the way up to a n times sigma n squared. Some special cases are of real interest here.

Now, Tejas, what if all the a i's were equal and equal to 1 and b was equal to zero? What would be capital Y? In this case, capital Y would be the sum of all the x i's. And then, if all the xi's were iid, that is, independent and identically distributed as normal with mean mu and variance sigma square, then what would be the expectation of this y?

It would be the sum of individual expectations. which is n times mu. That is the sum has a normal distribution with a mean equal to n times the mean of the individual xi's. But what do we know about the variance of y in this case? Applying the special case to the general formula, we get the variance of Y to be equal to n times variance of Xi or n times sigma square.

So the standard deviation will be the square root of n times the individual standard deviations. So when you add up the random variables, the standard deviations do not add up directly, the means add up and the expectation of y is n times mu. But the standard deviation of y is the square root of n times sigma i or the standard deviation of the individual x i's.

Let us look at another special case. Now, Tejas, what if all the a i's were equal and equal to 1 over n and b is equal to 0? What would be capital Y in this case? In this case, capital Y would be the sum of all the x i's divided by n, the average of n normally random variables, x i's. Yes, so let's denote the special case of y as x bar, the average of these xi's.

Then if all the xi's were iid, that is independent and identically distributed as normal with mean mu and variance sigma square, then what is the expectation of y? It would be the sum of individual expectations divided by n. which is n times mu divided by n, which is mu again. Yes, that is the average x bar has a normal distribution with the same mean as the individual xi's mu.

But what do we know about the variance of y or the variance of x bar in this case? Applying the special case to the general formula, we get the variance of y or the variance of x bar to be

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equal to sum of 1 over n square times variance of x i. n times sigma square divided by n square, that is sigma square divided by n.

Notice that averaging reduces the variance to sigma squared divided by n so that the average x bar of the individual xi's tends to get closer and closer to the mean value mu of the individual xi's. This tendency increases as n increases. That is, we are taking the average of more and more random variables, xi. Can we see an example?

Yes, that's what we're going to do next. Pankaj Vaghela in Jamnagar runs a machine shop. He manufactures pistons and cylinders that are used in automobile and oil and gas industries. He has a quality problem with the pistons and cylinders that are being manufactured in his shop. These are machined on multi axis automated machining centers. However, their dimensions are not always exact. From the machining data, he knows that the radius of the piston has a mean value of 30 millimeters and a standard deviation of 0. 05 millimeters. The inside radius of the cylinder into which this piston must fit has a mean value of 30. 25 millimeters and a standard deviation of 0. 06 millimeters. Data suggests that the radii of the piston head and the cylinder both have a normal distribution with the means and standard deviations that were just given. We can also assume that the machining of the pistons and the cylinders are independent. He wants to know whether given this variation in the dimensions, he will have cases where the piston may not fit into the cylinder.

Let's try and answer this. We first define the radius of the piston head as a normal random variable x1 with mean 30 millimeters and standard deviations equal to 0. 05 millimeters. We define the inside radius of the cylinder to be a normal random variable x2 with mean 30. 25 standard deviation 0. 06. Then the gap between the piston head and the cylinder can be defined as another random variable capital Y which is equal to X2 minus X1. Tejas, what would be the expectation of Y? The expectation of Y is going to be equals to 30. 25 minus 30 which is 0. 25. And what would be the variance of y? Given that x1 and x2 are independent, the variance of y would be the sum of the individual variances, 0. 05 squared plus 0. 06 squared, which comes out to be 0. 0061. Correct. Now, y will be normally distributed with a mean and computed. That is a mean of 0. 25 and a variance of 0. 0061. Then, the probability that the

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piston head will not fit within the cylinder would then correspond to the probability that this gap, y, takes a value less than or equal to 0.

Tejas, can you find this for us? Let's go to our normal template and update the value of the mean and the standard deviation and read off the probability that we need, we input the value of mean to be 0.25 and the standard deviation to be 0.078, and we can calculate the probability of Y less than or equal to zero in the cumulative distribution function section to be 0.0007.

So, Pankaj relax, there is a very small probability that the piston does not fit into the cylinder. Pankaj also believes that the gap must be between 0. 1 millimeter and 0. 35 millimeters for the system to perform optimally. What is the chance that this is likely to happen? The probability that the piston performs optimally would then be given by the probability that y takes a value between 0. 1 and 0. 35. Can you find this, Ashwini? We enter the value of capital A equals to 0. 1 and capital B equals to 0. 35 and we get the probability to be 0. 872. Yes, so Pankaj, Pankaj, No need to worry at all. The piston is most likely going to fit into the cylinder and there is more than an 87 percent chance that it will function optimally.

Chirag Heera is a farmer in Rajkot and he has a plantation where he grows cotton. Cotton plants during their initial few weeks after sowing grow to a height of about 20 to 30 centimeters. Chirag has observed that the height of a cotton plant after two weeks has a normal distribution with a mean of 29. 4 centimeters and a standard deviation of 2. 1 centimeters. He has planted 20 cotton plants and he wants to know the 95 percentile of the average height of the cotton plants after two weeks of growth. This will help him plan for the required irrigation. We can model the height of an individual cotton plant, Xi, as having a normal distribution with a mean mu i equal to 29. 4 centimeters and variance sigma i squared to be equal to 2. 1 square or 4. 41 centimeter square. Then, the average height of these cotton plants that Chirag has planted can be denoted by another random variable capital Y equal to X bar. The distribution of X bar is also normal with a mean of 29. 4 centimeters, but with a variance given by sigma i squared divided by 20 which is 0.2205. So the average height x bar would be normally distributed random variable with a mean of 29. 4 and a variance of 0. 2205.

We can now compute the 95th percentile for Chirag. We can go to the normal template and use the inverse section of the template to provide the answers for Chirag. We update the mean to

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be 29. 4, the standard deviation to be 0. 469. From the cumulative distribution section, we can find out that the height of the cotton plants would be less than 30. 17 centimeters 95 percent of the time. From the tail probability part, we can tell Chirag that 95 percent of the time, the average height of the cotton plant is going to be more than 28. 63 centimeters. We can also tell him that 95 percent of the time, the average height of these cotton plants is going to be between 28. 47 and 30. 32. Notice that the standard deviation of the height of the cotton plant is 0. 47 centimeters, which is considerably smaller than the standard deviation of the individual cotton plant's height, which was 2. 1 centimeters.

Now let's go back to Munaf Khan in Baruch, who has a different investment problem for us. Munaf Khan is now interested in investment that involves two kinds of stocks. Stock of a company A, whose returns are normally distributed with a mean of 8 percent and a standard deviation of 1. 5%. In addition, he is also interested in investing in stock of company B whose returns are also normally distributed, but with a mean of 9. 5 percent and a standard deviation of 2%. He considers the performance of both of these stocks to be independent. If you recall, he considers an investment to be excellent if the investment gives more than 10 percent returns. And he considers an investment to be unsatisfactory if it gives less than 5 percent returns.

And if the returns lie between 5 % and 10%, then the investment is considered to be moderate. Munaf wants to know what is the probability that the stock of company A would be moderate. First, let's denote the returns of stocks A and B by random variables XA and XB, which have the mean and standard deviations that Munaf has given us.

And if the returns from the stock of company A is moderate, then XA has to take a value between 5 and 10%. We use our normal distribution template and update the mean to be 8 and the standard deviation to be 1. 5 and we put in the value of a and b to be 5 and 10 respectively. What do you get Tejas? I get the probability that the stock performs moderately to be 0. 886. Yes, and next, Munaf wants to know the probability that the performance of the stock of company B would be excellent. The stock of company B will be excellent if it gave us a return greater or equal to 10%. So update the template to reflect the mean and standard deviation of the stock of company B.

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And we read off the tail probability corresponding to 10. What do you get, Ashwini? I get it to be 0. 4013. Yes. And next, Munaf wants to know the probability that the stock of company B performs better than the stock of company A. To compute this, we define a new random variable, capital Y which is equal to XB minus XA.

Then, we know that the random variable Y will also have a normal distribution, but with mean equal to 9. 5 minus 8, which is 1. 5. And the variance of Y would be equal to the sum of the variances, 4 plus 2. 25, which is 6. 25. Therefore, Y would be normally distributed with a mean of 1. 5 and a standard deviation of 2. 5. Munaf now wants to know whether the stock of company B would perform better than the stock of company A. So we need to find the probability that Y takes a value greater or equal to zero. We go to our normal template. And we update the mean and the standard deviation. And then we read the tail probability corresponding to zero for y.

What do you get, Tejas? I get that to be equal to 0. 726. So there is a 73 percent chance that the stock of company B will outperform the stock of company A. But now, Munaf also wants to know what is the probability that the company B stock performs at least 2 percentage points better than the stock of company A. In this case, we want the probability of Y being greater or equal to 2.

We compute the tail probability corresponding to 2 and we get that to be equal to 0. 421. So, there is a 42 percent chance that the stock of company B will be 2 percentage points higher than the stock of company A. So far, we have seen several examples related to the normal random variable, the standard normal random variable, and linear combinations of normal random variables.

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Chi-Square Distribution

Before we close this module, we will take a preview of some of additional continuous distributions that are related to or derived from the normal distribution. These distributions will be very useful when we are looking at statistical inference and its applications later in the course. The purpose of this preview is one, to introduce you to these distributions, two, to share some properties of these distributions, and three, to see how they are related to the normal distributions. We will solve detailed examples of these applications as part of the statistical inference techniques that we will cover later in this course.

A notation alert though, some of these distributions have strange names and complicated probability density functions. And ugly integrations, I assume. Possibly, but we will not have to do the integration. Templates to the rescue again. Yes, of course. We will be covering three particular distributions. The first distribution we will be covering is the chi-square distribution. The next distribution we will be covering is the student t distribution. And the third distribution that we will be covering is known as the f distribution.

For each of these distributions, we will see the probability density function. We will see the relationship to the normal distribution. And like we did for the standard normal, we will do the critical value calculations on the tail probabilities of each of these distributions. Let's get started.

If a random variable Z has a standard normal distribution, then the random variable defined by X is equal to Z square is said to have a chi-square distribution with one degree of freedom.

More generally, if several random variables Z i are all independent standard normals for i equal to 1 to n, then the random variable X obtained by their sum, that is the sum of Z 1 square plus Z 2 square, all the way up to Zn square is said to have a chi-square distribution with n degrees of freedom.

Where is the chi-square distribution used? The chi-square distribution plays a pivotal role in analytics. It is used in many non-parametric hypothesis tests. It forms the basis of several

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feature selection and classification techniques. It is also used in logistical regression models to test independence of variables.

What is degrees of freedom in a chi-square distribution? The degree of freedom in a chi-square distribution corresponds to the number of squares of standard normal random variables that are being added. What are some of its properties?

First, distribution is defined only for positive values of the random variable. The second feature is that the chi-square distribution is a single parameter distribution. The parameter of the distribution corresponds to the degrees of freedom. The mean of the chi-square distribution is equal to the degrees of freedom. The variance of the chi-square distribution is equal to two times the degrees of freedom.

Next, as the degrees of freedom increases, the probability density function of a chi-square distribution converges to a normal distribution. This is the result of the center limit theorem that we will see a little bit later. And finally, the chi-square distribution is used in the goodness of fit hypothesis step.

This test is one of the more popular hypothesis tests that is used to check whether a data set follows a particular distribution. The probability density function looks messy. Yes, it is a bit messy, but in Microsoft Excel, the function chi-square dot dist can be used to calculate the cumulative distribution of a chi-square distribution with k degrees of freedom.

In Excel, this function has three parameters. The first parameter is x, that corresponds to the value of the chi-square random variable for which we are trying to compute the probability. The next parameter is k, which corresponds to the degree of freedom of the chi-square random variable that we are trying to analyze.

And the third parameter is a toggle between true or false, where true gives us the cumulative distribution function and false gives us the probability density function. Here we have a template where this function has been used. This template works in a similar manner like the template for the normal distribution.

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Once we put in the degrees of freedom in cell E3, the mean and the standard deviation are computed in cells B3 and C3 respectively. Then we have sections for the probability calculation and inverse calculation like we had for the normal distribution. On the right hand side, we also have the plot of the probability density function for the chi-square distributions with various degrees of freedom.

The critical points of a chi-square distribution require a definite mention at this point, because we are going to see them later. They are denoted by chi-square alpha comma k. It is defined as the value of the random variable for which the tail probability is equal to alpha. In other words, the probability that a chi-square random variable X with k degrees of freedom is greater or equal to the critical value chi-square alpha comma k is equal to alpha.

Let us now look at various values of alpha levels of 0. 2, 0. 1, 0. 05, and 0. 01, and compute the critical values corresponding to the various degrees of freedom of k. k equal to 5, 10, 15, and 20. Let's compute the critical points for chi-square distribution denoted by chi-square alpha comma k. Let's vary alpha to take values of 0. 2, 0. 1, 0. 05, and 0.01. These all correspond to tail probabilities. Then, the critical values corresponds to the value of the random variable for which the right tail has these probabilities. Let us consider a chi-square random variable with 5 degrees of freedom. We compute these critical values using our template.

So we input the degrees of freedom to be equal to 5 in cell E3. Notice that the mean and the standard deviations get updated in cell B3 and C3 respectively. Then we go to the inverse section of the template and put in the values of the probability that we are trying to analyze, namely 0. 2, 0. 1, 0. 05, and 0. 01 in the tail probability section. Then we get the critical values of the chi-square distribution to be equal to 7. 3, 9. 2, 11. 1, and 15. 1, respectively. Notice that this corresponds to the case where the degree of freedom is equal to 5. Ashwini, can you repeat this calculation for degrees of freedom equal to 10?

And Tejas, can you repeat this for the case where the degrees of freedom is equal to 15? Yes, when I change the degrees of freedom to be equals to 10, I get the critical values to be equals to 13. 4, 16. 0, 18. 3, and 23. 2. And Tejas, what did you get? I got 19. 3, 22. 3, 25, and 37. 6. Finally, if we do it for degrees of freedom equal to 20, then we get the values of 25, 28. 4, 31. 4, and 37. 6.

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A few points to note. First, the computation of critical values is fairly easy. It is just the values corresponding to the tail probabilities. Unlike the standard normal, where we had only one critical value corresponding to the values of alpha, here we could get different critical values corresponding to the degrees of freedom of the chi-square distribution.

Notice also, as the degrees of freedom starts to increase, the critical values also start to increase. That's because the distribution has a higher standard deviation and is flatter. Additionally, you will also notice that the skewness of the chi-square distribution starts to reduce as the degrees of freedom starts to increase and it starts to look more and more symmetric and closer to the bell shape as the degrees of freedom increase. This is all we need to know about the chi-square distribution for now. We will get into more details when we are using it in certain applications.

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Student's t-Distribution

Let's now look at the next distribution related to the normal distribution, which is the T distribution. What is a T distribution? And hold on Tejas and Ashwini, this is going to get messy. If a standard normal random variable Z is divided by the square root of an independent chi square random variable with k degrees of freedom divided by k.

Then the resulting random variable is said to have a t distribution with k degrees of freedom. Oh no, z divided by the square root of chi-square. Yes, the formal definition of a t random variable with k degrees of freedom is that a standard normal random variable in the numerator divided by the square root of a chi-square random variable with k degrees of freedom divided by k in the denominator.

This distribution is often referred to as the student's t distribution with k degrees of freedom. Tejas and Ashwini, can you guess why it might be called a student t distribution? I think some European named student wanted to make our lives miserable. Ha ha, you're partly correct. Actually, there is a very interesting story.

There is always an interesting story behind someone's misery. The tea distribution was discovered by William Sealy Gosset he studied mathematics and chemistry at the Oxford University in England. In 1899, he moved to Dublin in Ireland, where he worked as a statistician for the Guinness Brewery. During his work on the quality of barley and hops, William Gosset proposed the use of the tea distribution. In 1908, He published his ideas in an academic article using the pseudonym Student, because Guinness did not allow its employees to publish their own research results. Consequently, the t distribution is often referred to as the student's t distribution. I do not know what to say. If I got an internship at Guinness, the last thing I would be doing is working on a statistics distribution.

Tejas, if it wasn't for Sealy Gosset, we would be really in trouble in many applications. What are the applications of the student's t distribution? First, it is used in a lot of hypothesis tests about means of a population especially for small sample sizes. The student distribution is also used in hypothesis testing and in the diagnostics of linear regression models.

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If you look at the plot of a student t distribution, you will see that it has the shape that is very similar to that of a standard normal distribution. It has a symmetric bell shaped curve and it is centered at zero but it is actually a little flatter than the standard normal distribution. However, as the degrees of freedom k increases, the t distribution starts to look closer and closer to a standard normal distribution.

The standard normal distribution is in fact the limiting value of a t distribution as the degrees of freedom k goes to infinity. What are some of the properties of the t distribution? First, the mean of a t distribution with two or more degrees of freedom is zero, like the standard normal. However, the variance of the t distribution with k degrees of freedom, where k is greater than two, is given by k divided by k minus two.

As the degrees of freedom increases, the probability density function of a t distribution converges to that of the standard norm. The t distribution is an important distribution for the hypothesis test of means of a population. and for comparing the means of two populations. If the chi-square distribution has a messy PDF, I guess the pdf of the t distribution would be more messy.

Yes, Ashwini, the form of the pdf is a bit complicated, but by now you should realize that we mostly care about the critical values, the tail probabilities, and the cumulative probabilities. And as long as we can compute The exact functional form does not bother us much. Are you saying we have a template for the t distribution as well?

Yes, Excel has inbuilt functions for the t distribution as well. Oh, thank you so much Excel. What would I do without you? What are these inbuilt functions? The cumulative probabilities of the student t distribution for various values of degrees of freedom can be computed in Microsoft Excel using the in built function t.dist. This function requires three parameters. The first parameter is x, which corresponds to the value of the t random variable for which we are trying to compute the probability. The second parameter is k which corresponds to the degrees of freedom of the t distribution that we are analyzing. And the third parameter is a toggle that takes a value of true or false.

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If the toggle takes a value of true, then we get the cumulative probability corresponding to the other parameters that we have chosen. If the toggle is set as false, then we get the probability density function corresponding to the other parameters that we have chosen. Let us now talk a little bit about the critical values of the t distribution with k degrees of freedom.

Notice that the critical value computation would follow a similar pattern like we did for the standard normal. Like the standard normal distribution, the t distribution is a symmetric bell shaped distribution. It is centered at zero, but flatter. It has degrees of freedom k. We define the critical value known as T alpha by 2, k. This is the value of the T random variable that has a tail probability of alpha by 2. By symmetry, the cumulative probability of minus T alpha by 2 comma k will also be alpha by 2. The area or the probability to the right of T alpha by 2 comma k, that is the tail probability corresponding to T alpha by 2 comma k, is equal to alpha by 2.

Similarly, the value of minus T alpha by 2 is that value of t such that the cumulative probability of minus t alpha by 2 comma k is equal to alpha by 2. Therefore, the probability that t lies between minus t alpha by 2 and plus t alpha by 2 is going to be equal to 1 minus alpha. Let us now compute the critical points of a t distribution denoted by t alpha by 2 comma k.

Let's vary alpha by 2 to take values of 0. 1, 0. 05, 0. 025, and 0. 005, like we did for the standard normal. We know that these correspond to tail probabilities. Then the critical values corresponds to the value of the random variable for which the right tail has these probabilities. Let us consider a t random variable with 5 degrees of freedom.

We compute these critical values using our template. So we input the degrees of freedom to be equal to 5 in cell E3. Notice that the mean and the standard deviation get updated in cells B3 and C3 respectively. Then we go to the inverse section of the template and we put in these values of the probabilities that we are trying to analyze namely 0. 1, 0. 05, 0. 025, and 0. 005 in the tail probability section. Then we get the corresponding critical values of the T random variable for these tail probabilities to be equal to 1. 48 2. 02, 2. 57, and 4. 03 respectively. Note that this corresponds to the case where the degree of freedom is equal to 5.

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Ashwini and Tejas, can you repeat these calculations for degrees of freedom equal to 10 and 15? When I change the degrees of freedom to be equals to 10, I get the critical values to be equals to 1. 37, 1. 81, 2. 23 and 3. 17 respectively. Yes, and Tejas, what did you get for degrees of freedom equal to 15? I got 1. 34, 1. 75, 2. 13 and 2. 95. Excellent, and finally if we do it for degrees of freedom equal to 20, we get the values of 1. 33, 1. 73, 2. 09 and 2. 85 respectively. A few points to note. First, the computation of critical values is fairly easy for the t distribution as well. In fact, it is very similar to what we did for the standard normal.

The critical values are similar too. Yes, they are. And in fact, you will notice that these values decrease as the degrees of freedom increase. This is because as k increases, the t distribution is less flat and it starts to converge to a standard normal distribution. And if you compare the critical values with those of the standard normal, you will see that our critical values tend to those values as we increase k.

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F distribution

Let's now look at the F distribution. The F distribution is a short form for Fisher distribution and is named after the famous statistician Ronald Fisher. It is the ratio of two independent chi-square random variables divided by their own respective degrees of freedom. Therefore, the F distribution can be written as F of k1, k2 which is the ratio of a chi-square with k1 degrees of freedom divided by k1 in the numerator divided by a chi-square with k2 degrees of freedom divided by k2 in the denominator. In the F distribution, the degrees of freedom k1 and k2 correspond to the degrees of freedom of the chi-square distributions.

k1 corresponds to the degree of freedom of the chi square distribution in the numerator, and k2 corresponds to the degree of freedom of the chi-square distribution in the denominator. In general, an F distribution with k1 degrees of freedom in the numerator and k2 degrees of freedom in the denominator is not the same as an F distribution with k2 degrees of freedom in the numerator and k1 degrees of freedom in the denominator.

What are some properties of the F distribution? The F distribution is also defined only for positive values of the random variable. It is unimodal and has a long right hand tail and it is skewed. The mean of the F distribution is given by k2 divided by k2 minus 2 for k2 values that are greater or equal to 2.

The standard deviation of the F distribution decreases as k2 increases. And also, as k1 and k2 become larger, the probability density function becomes more and more sharply spiked about the value of 1. Where are we going to see the F distribution? The F distribution is used quite heavily in the analysis of variance.

It also plays an important role in feature selection in many machine learning algorithms. It is also used in testing the overall fitness of multiple linear regression models. Do we have inbuilt functions and a template in Excel? Yes, Excel has inbuilt functions and we have a template. The cumulative probabilities of the F distribution for various values of k1 and k2 can be computed in Microsoft Excel using the inbuilt function f. dist. This function requires four parameters. The first parameter is x, which corresponds to the value of the F random variable for which we are trying to compute the probability. The second parameter is k1, which

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corresponds to the degrees of freedom of the chi square distribution in the numerator. The third parameter is k2 which corresponds to the degrees of freedom of the chi-square distribution in the denominator. And the fourth parameter is a toggle that takes a value of true or false. If the toggle takes a value of true, then we get the cumulative probability corresponding to the other parameters that we have chosen.

If the toggle is set at false, then we get the probability density function corresponding to the other parameters that we have chosen. In the template, these functions have been used to compute the probabilities as well as the inverse probability calculations. We enter the values of k1 and k2 in cells C3 and D3.

The mean and the standard deviations get updated accordingly. Let us now compute the critical points of an F distribution, denoted by F of alpha, k1, k2. Let us vary alpha to take values of 0. 2, 0. 1, 0. 05, and 0. 01. These correspond to tail probabilities. Then, the critical values corresponds to the value of the random variable for which the right tail has these probabilities.

Let us consider an F random variable with k1 equal to 5 and k2 equal to 5. We compute these critical values using our template. So we input k1 and k2 in cells C3 and D3. Notice that the mean and the standard deviation get updated in cells B3 and C3. Then, we put into the inverse section of the template the values of the probability that we are going to analyze, namely 0. 2, 0. 1, 0. 05, and 0. 01 in the tail probability section. Then we get the corresponding critical values of the F random variable for these tail probabilities to be equal to 2. 23, 3. 45, 5. 05, and 10. 97, respectively. Notice that this corresponds to the case where k1 is equal to k2 is equal to 5. Ashwini, can you repeat this calculation for k1 equal to 5, but k2 equal to 10?

And Tejas, can you repeat this for the case where k1 is equal to 10, but k2 is equal to 5? Yes. For k1 equals to 5 and k2 equals to 10, I get the critical values to be equals to 1. 80, 2. 52, 3. 33, and 5. 64. And Tejas, what did you get? I got 2. 19, 3. 30, 4. 74, and 10. 05. And finally, if we do it for k1 equal to 20 and k2 equal to 20, we get the values of 1. 47. 1. 79, 2. 21, and 2. 94, respectively. A few points to note here again. Notice that when we switch the values of k1 and k2, the critical values also change. So, we must be careful when we use the F distribution and ensure that we have the right values of k1 and k2 when we compute the critical values. A

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second point to note is that the values of k1 and k2, particularly when k2 increases from 5 to 10 to 20, we notice that the critical values start to reduce in magnitude.

In fact, when k1 and k2 are both equal to 20, these critical values are close to one. This is because of a point I made earlier, which is that as the value of K2 increases, the F distribution starts to peak and take values pretty close to one. This is all we need to know about the F distribution at this point.

So to summarize, we had a brief overview of three distributions related to the normal, namely the chi-square, the T distribution and the F distribution. As mentioned earlier, these distributions will come in handy when we start to look at statistical inference in the later part of this course. Let's summarize before we wrap up this module.

We covered a fair bit of ground in this module. We started with continuous distributions. We built on what we had done with discrete probability distributions in the previous module. We started with the fundamentals of continuous probability distributions using the notion of probability density function as opposed to probability mass function.

Then, we looked at two simple continuous distributions, namely the uniform and the exponential distribution. We looked at their properties and covered several examples illustrating their applications. Then, we looked at the normal distribution. We looked at both the general normal distribution with mean mu and standard deviation sigma, and the standard normal distribution with mean zero and standard deviation of one.

We saw the connections between the two distributions and several applications. We also looked at the computation of critical values corresponding to the standard normal distribution. We also looked at how the normal distribution is used to approximate discrete distributions such as the binomial and the Poisson.

Then, we looked at linear combinations of normal random variables and saw how they retain the normal characteristics. We saw some applications of these concepts. And finally, we looked at three distributions that are related to the standard normal, namely the chi square distribution, the t distribution, and the f distribution.

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We looked at their properties, their applications, and how to compute their critical values. In the next module, we will transition to statistical inference. We have now laid the groundwork for this transition. Thank you Tejas and Ashwini for your continued support. I will see you next time.

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